Radxa E54C

Schematic

V1.2

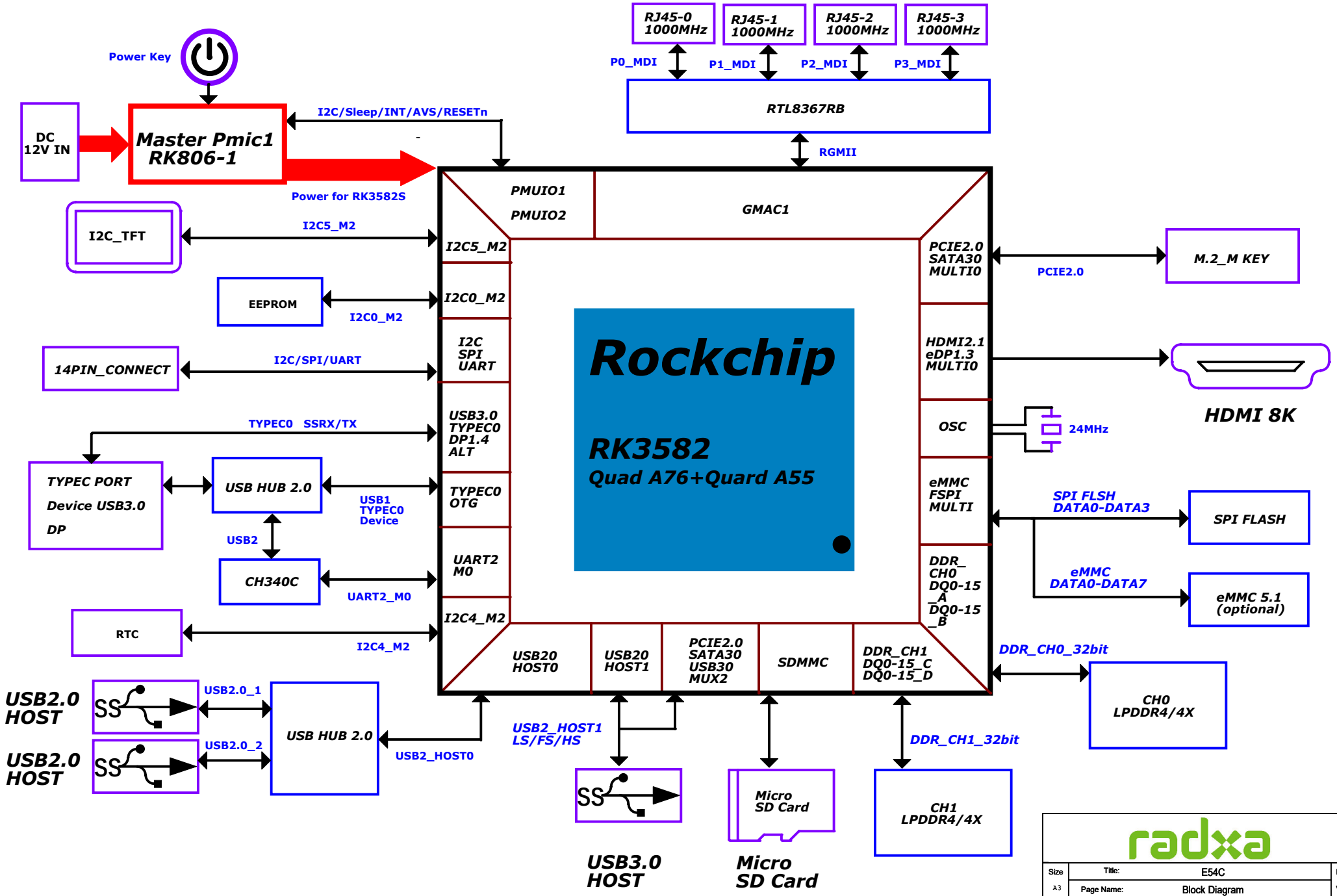
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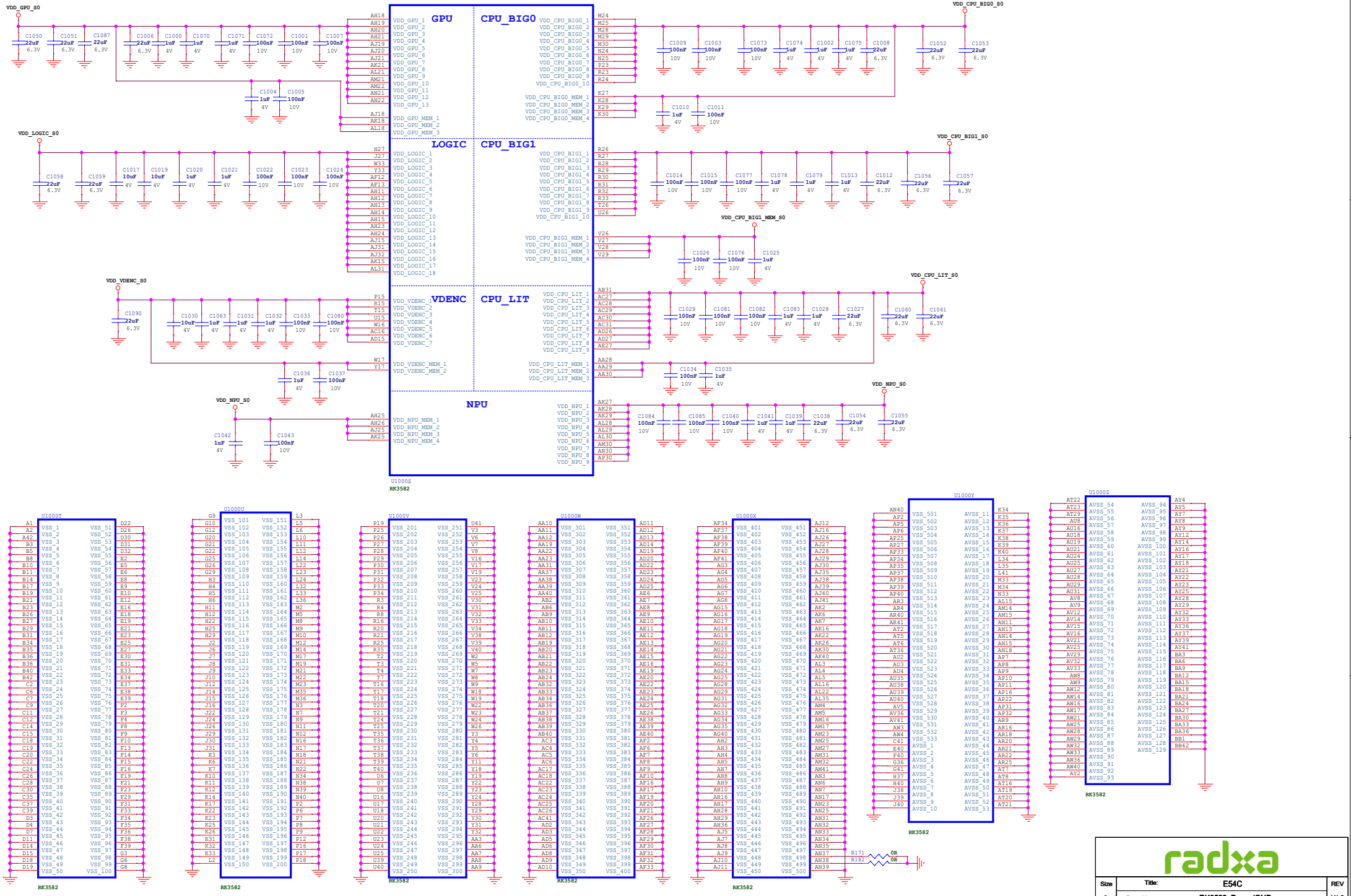
E54C Block Diagram



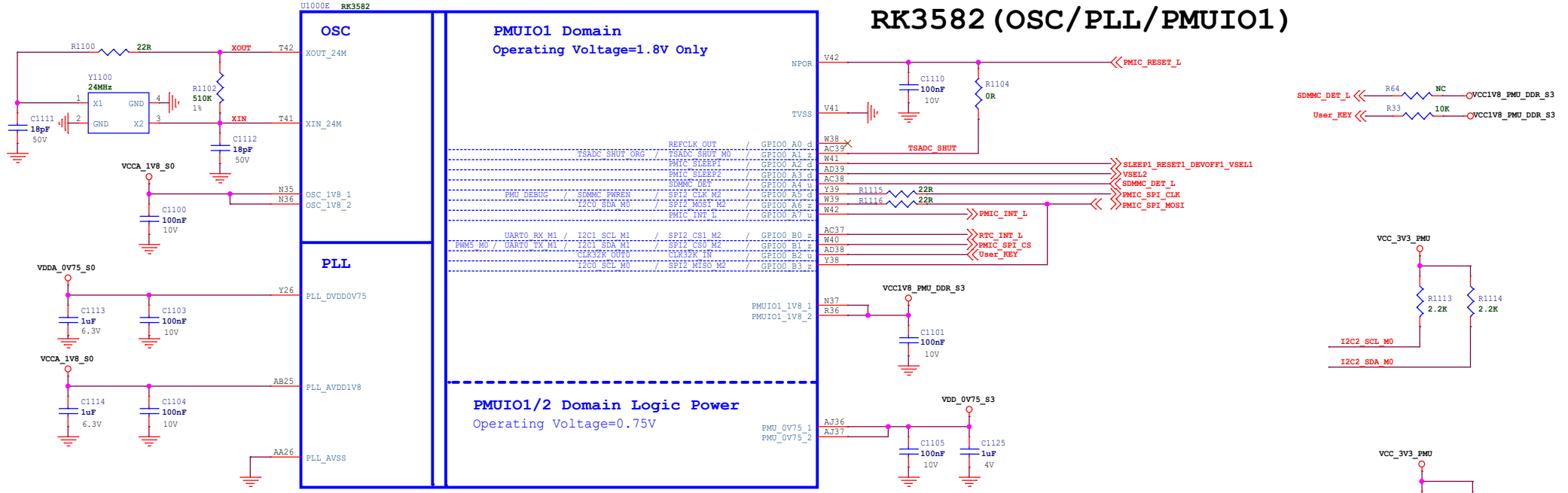
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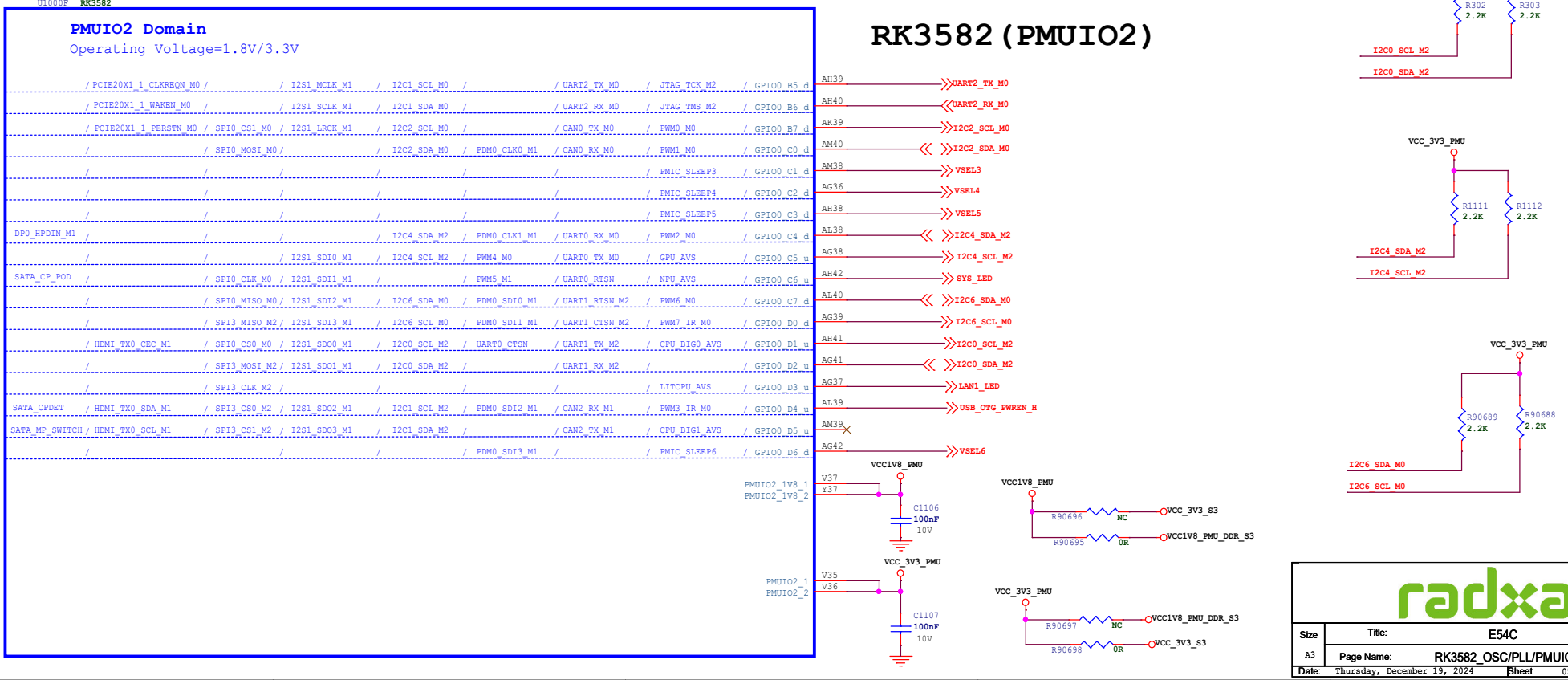
RK3582 (Power&Gnd)



RK3582 (OSC/PLL/PMUIO1)

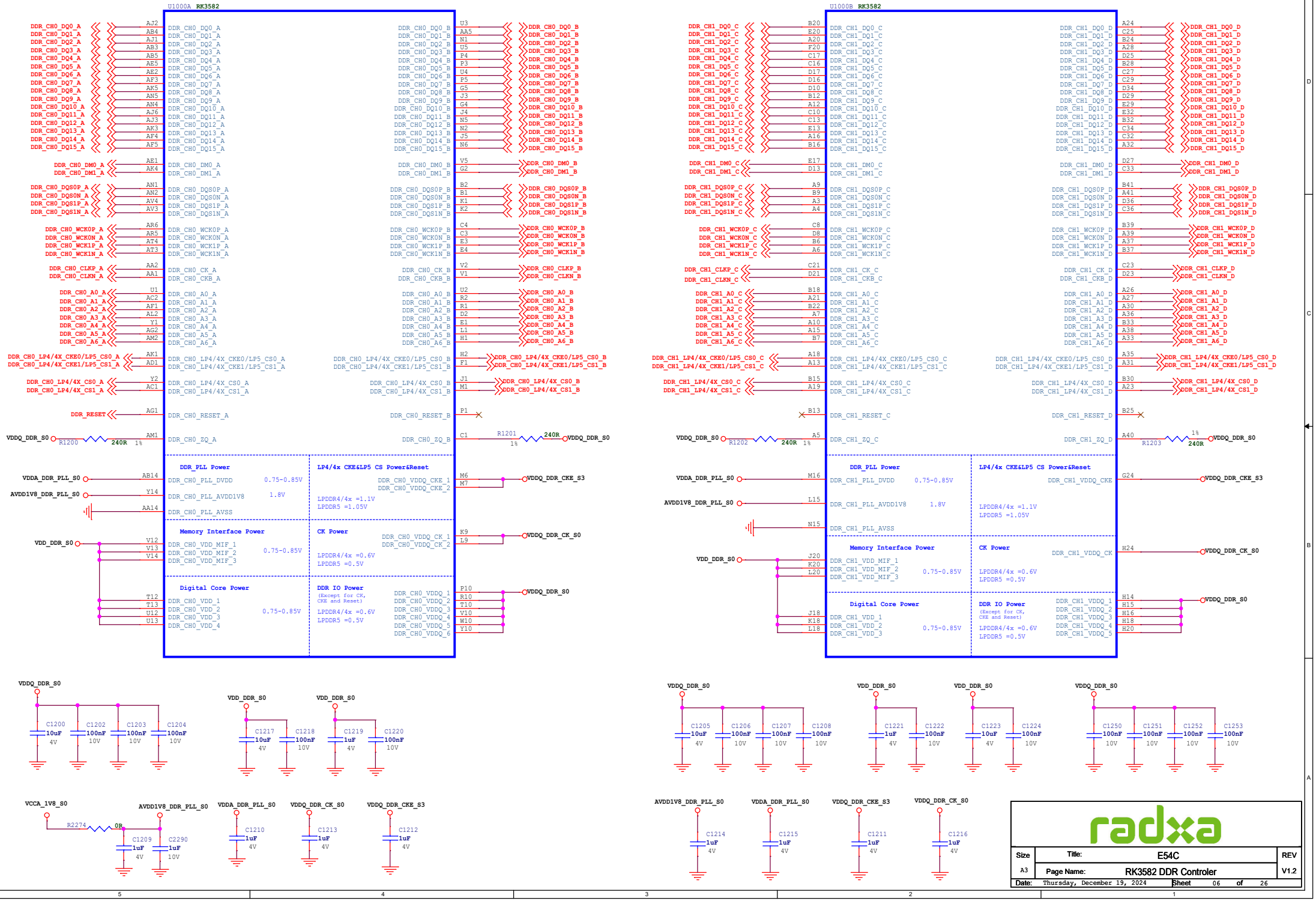


RK3582 (PMUIO2)



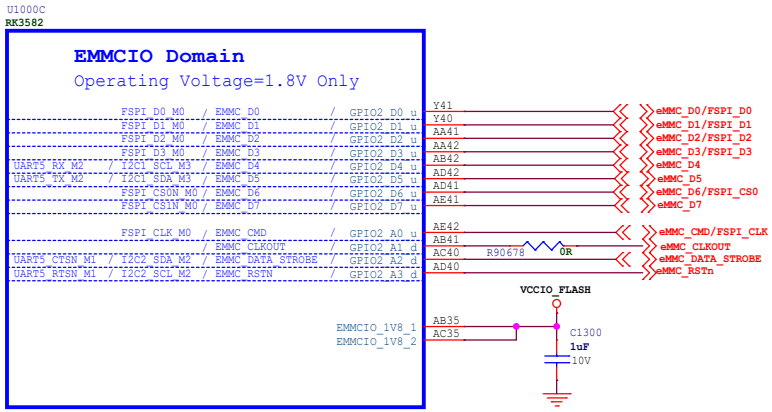
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RK3582 (DDR PHY)

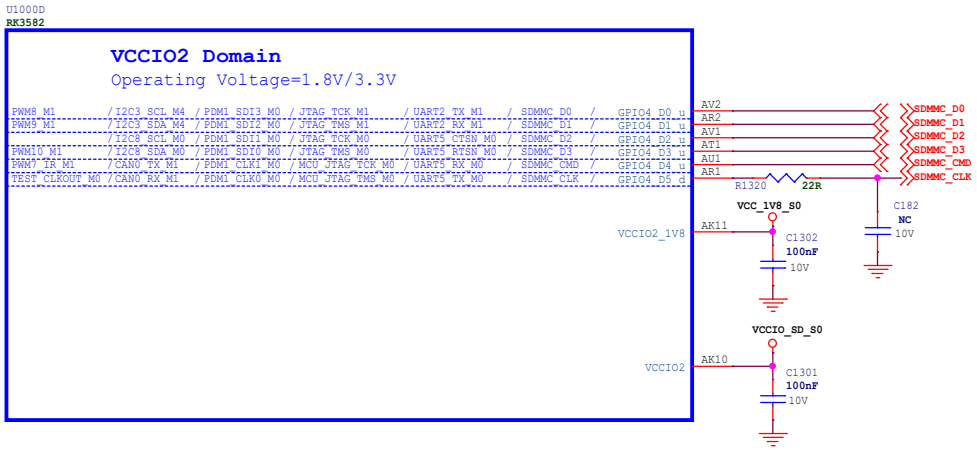


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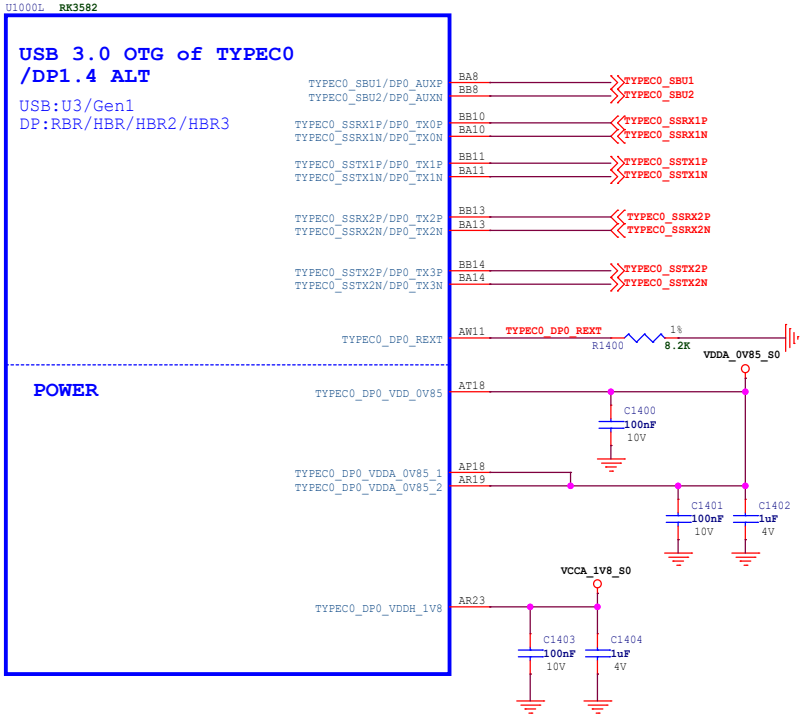
RK3582 (EMMCIO Domain)



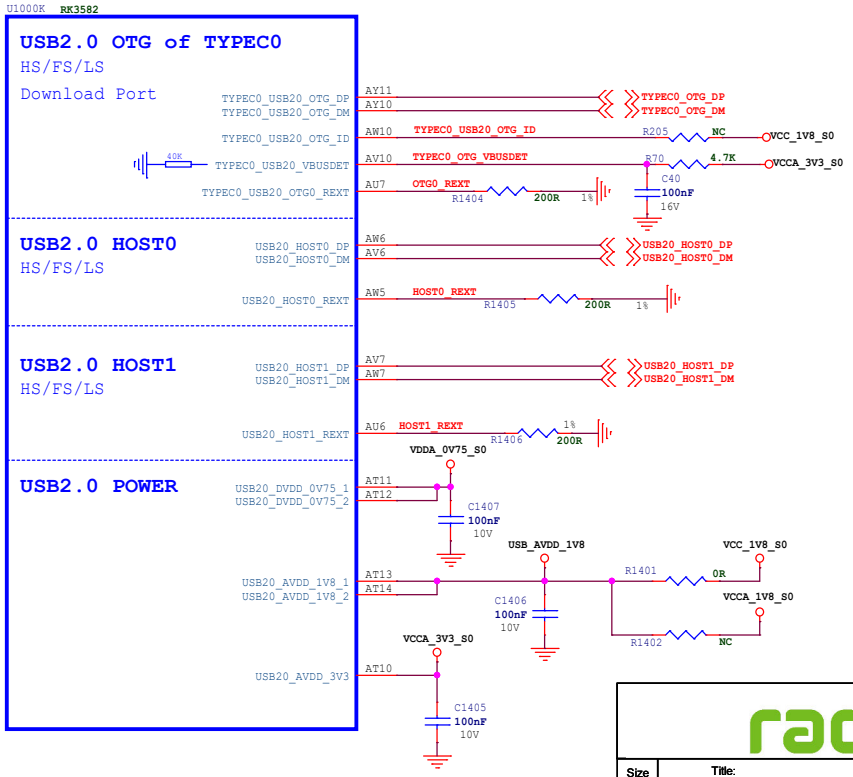
RK3582 (VCCIO2 Domain)



RK3582 (USB3.0/DP1.4)



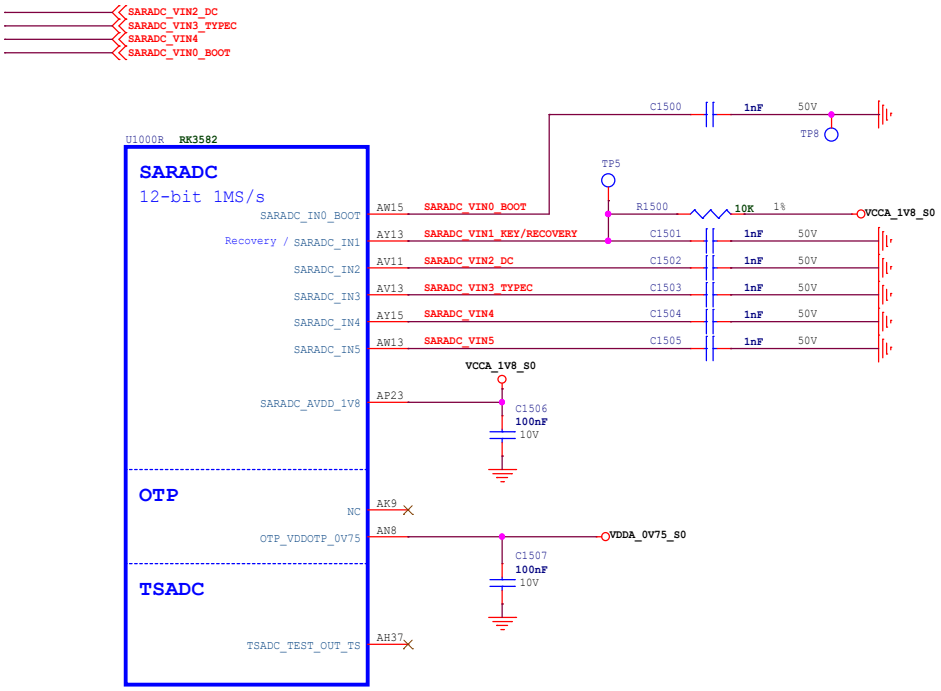
RK3582 (USB2.0)

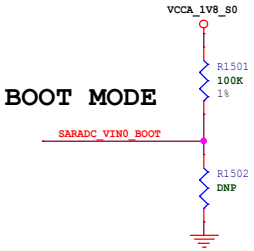


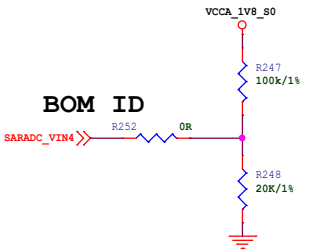
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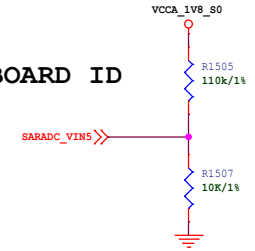
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RK3582 (SARADC/OTP/TSADC)

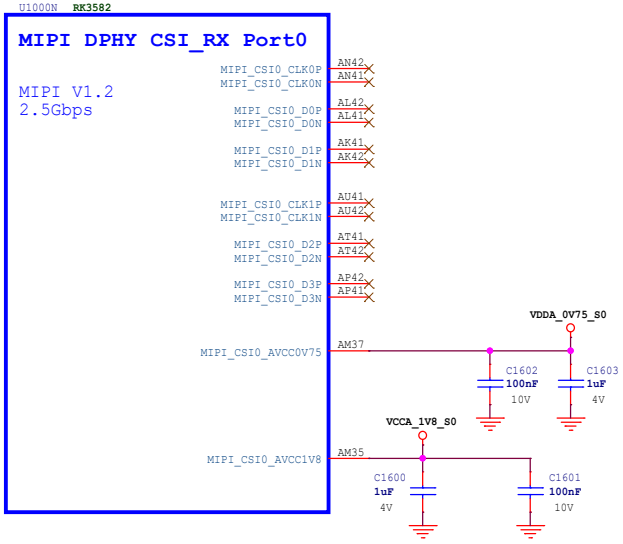




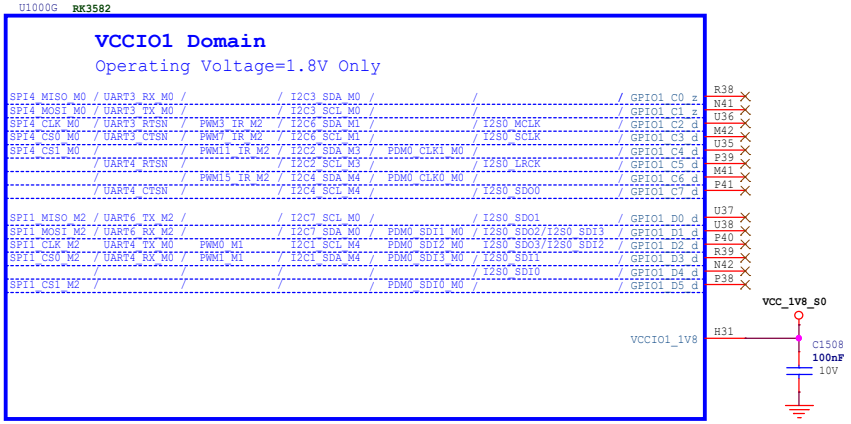




RK3582 (MIPI_DPHY CSI0 RX)

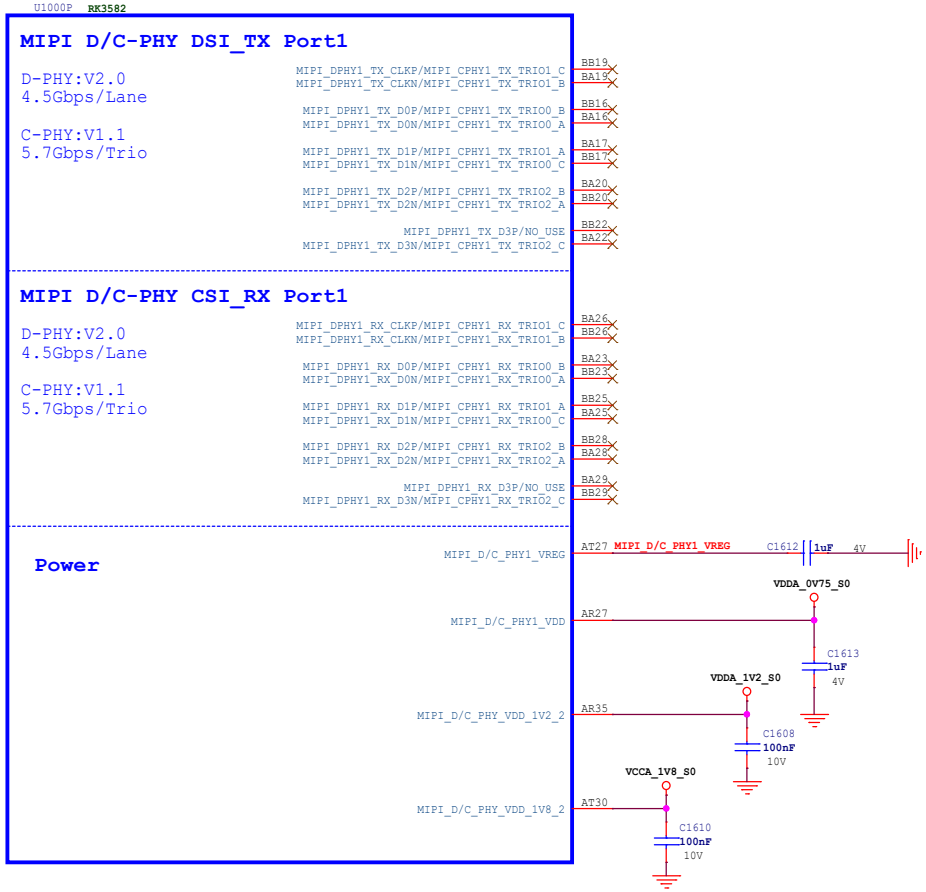


RK3582 (VCCIO1 Domain)

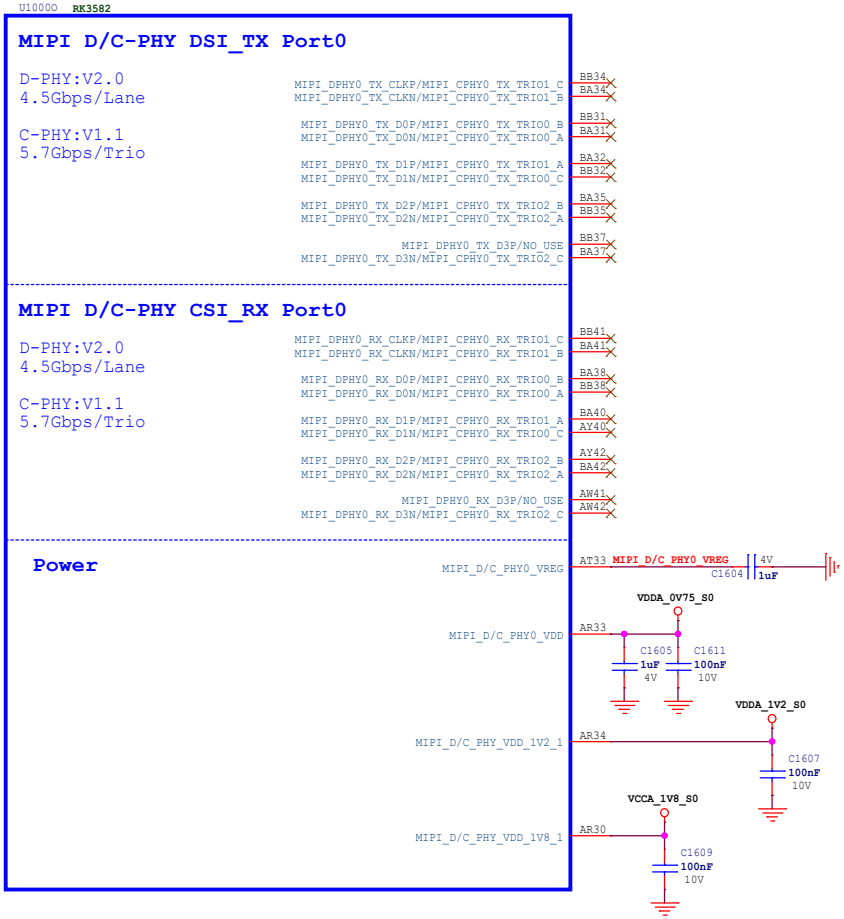


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RK3582(MIPI_D/C PHY1)

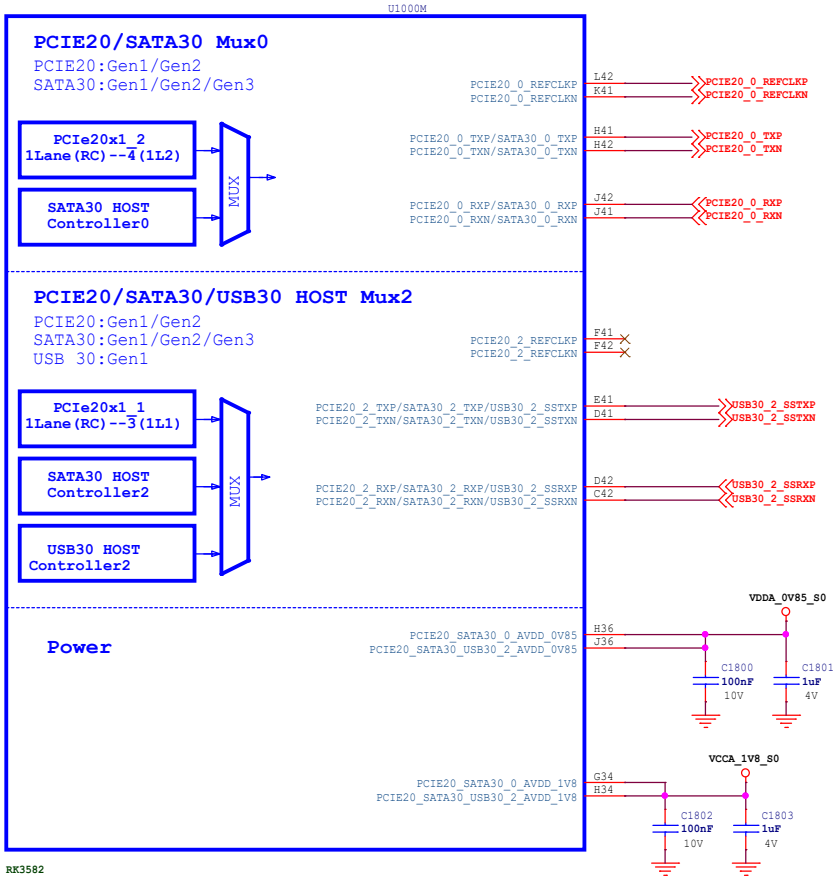


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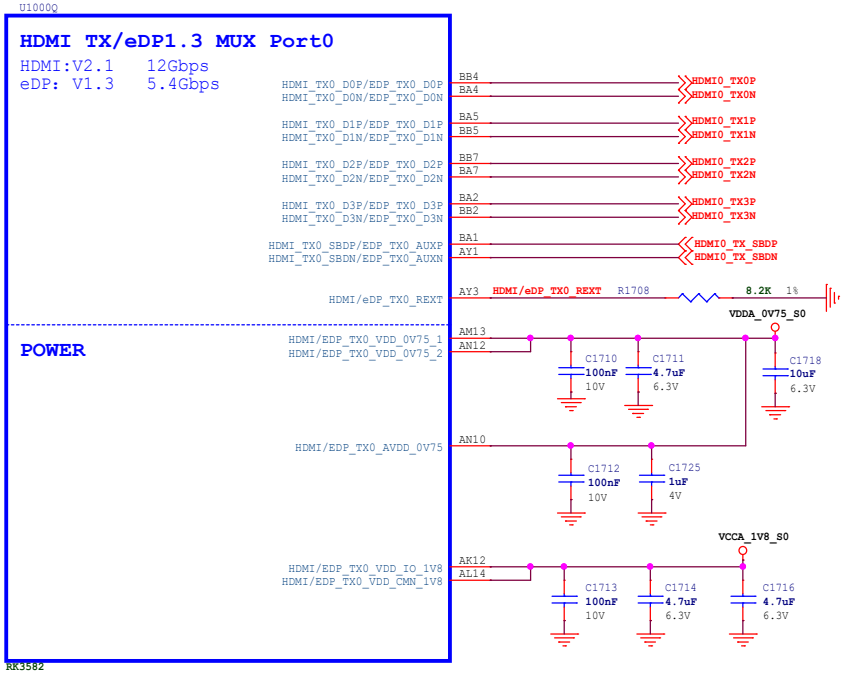


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RK3582(PCIE20/SATA30/USB30)

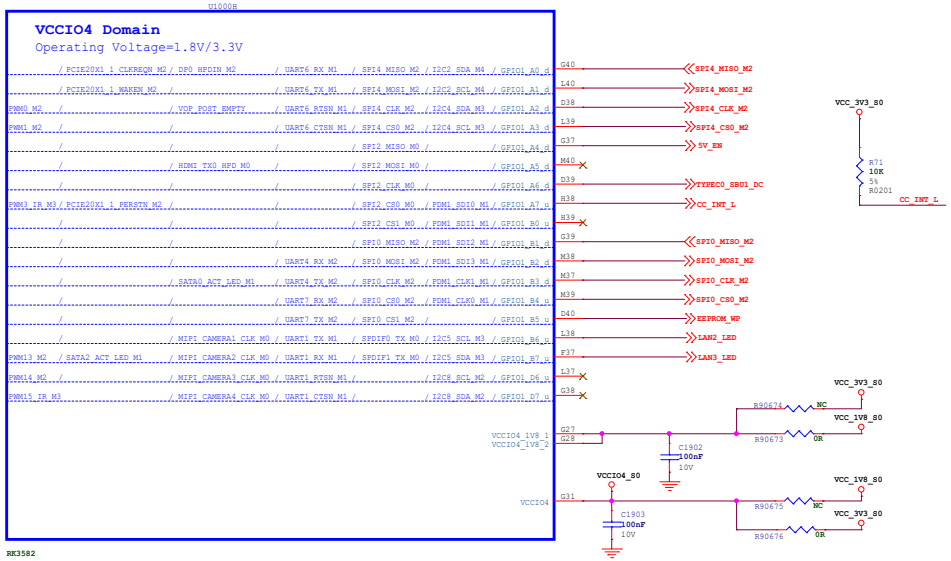


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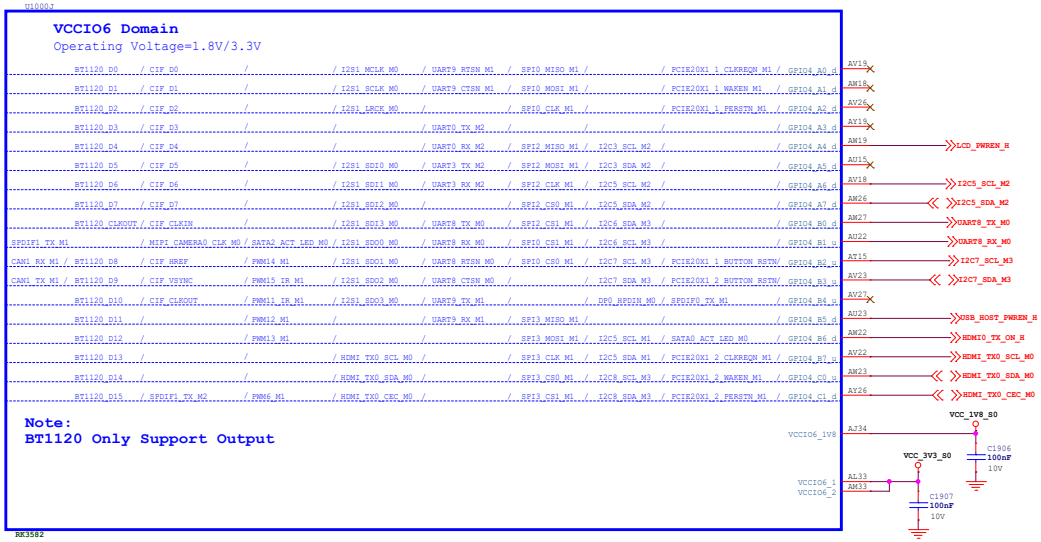


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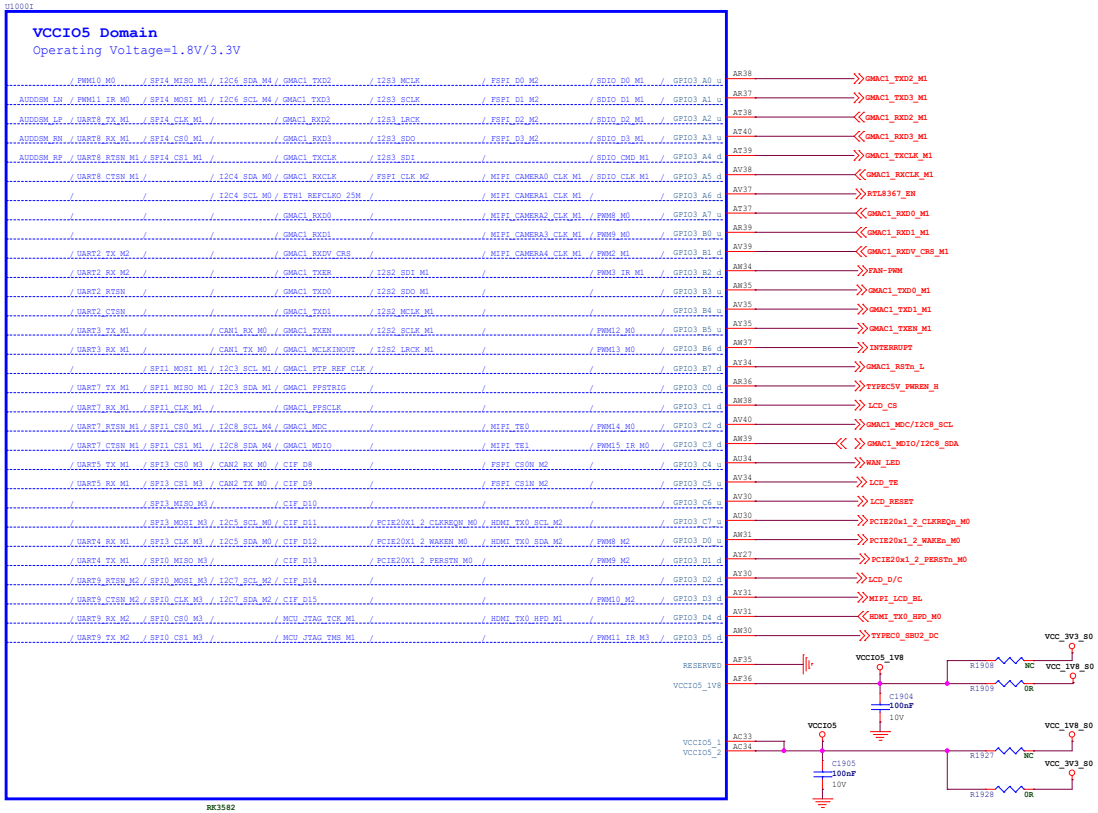
RK3582 (VCCIO4 Domain)

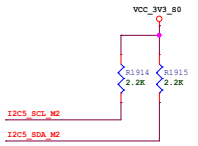


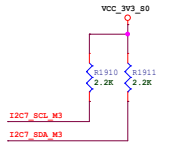
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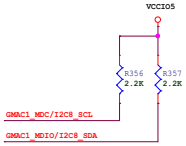


RK3582 (VCCIO5 Domain)





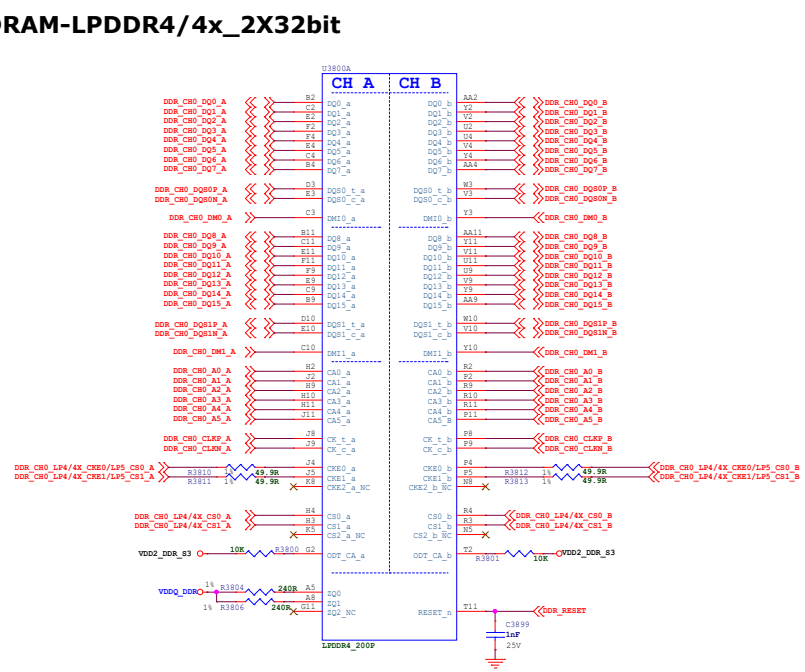


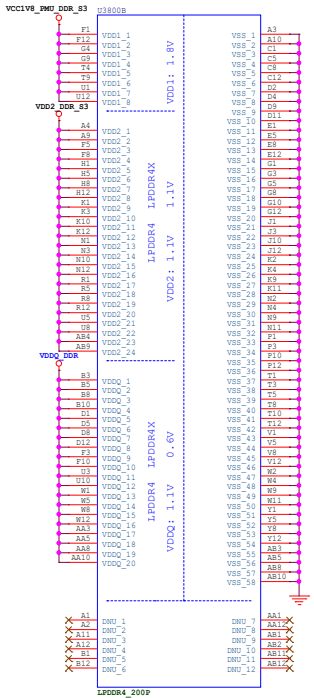


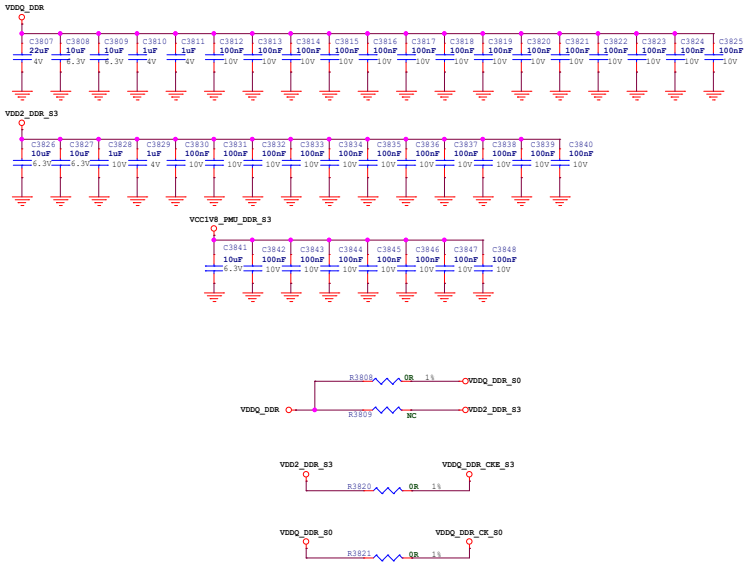
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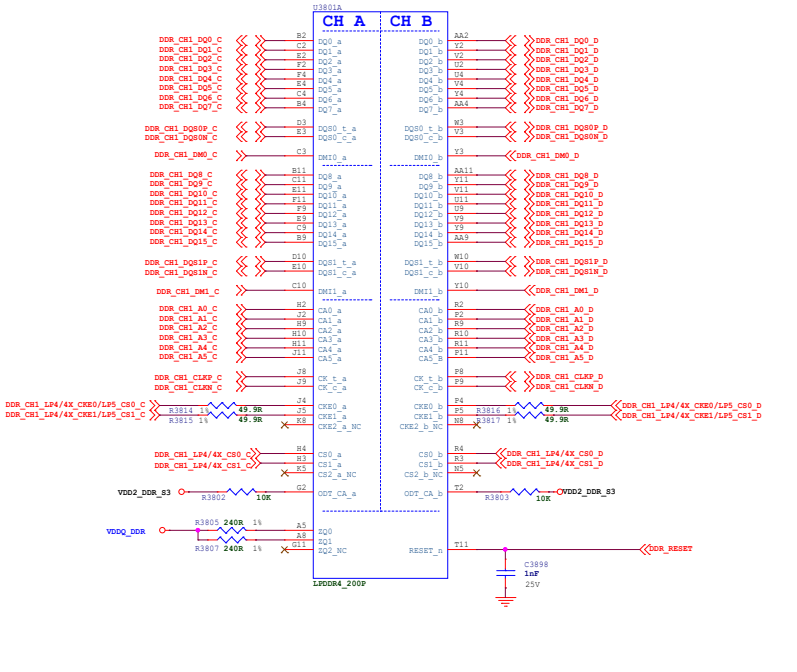
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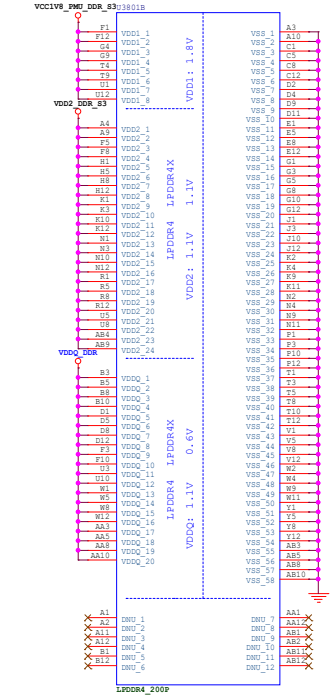
DRAM-LPDDR4/4x_2X32bit

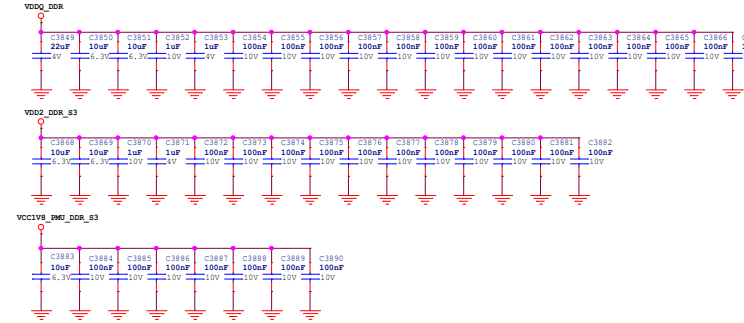






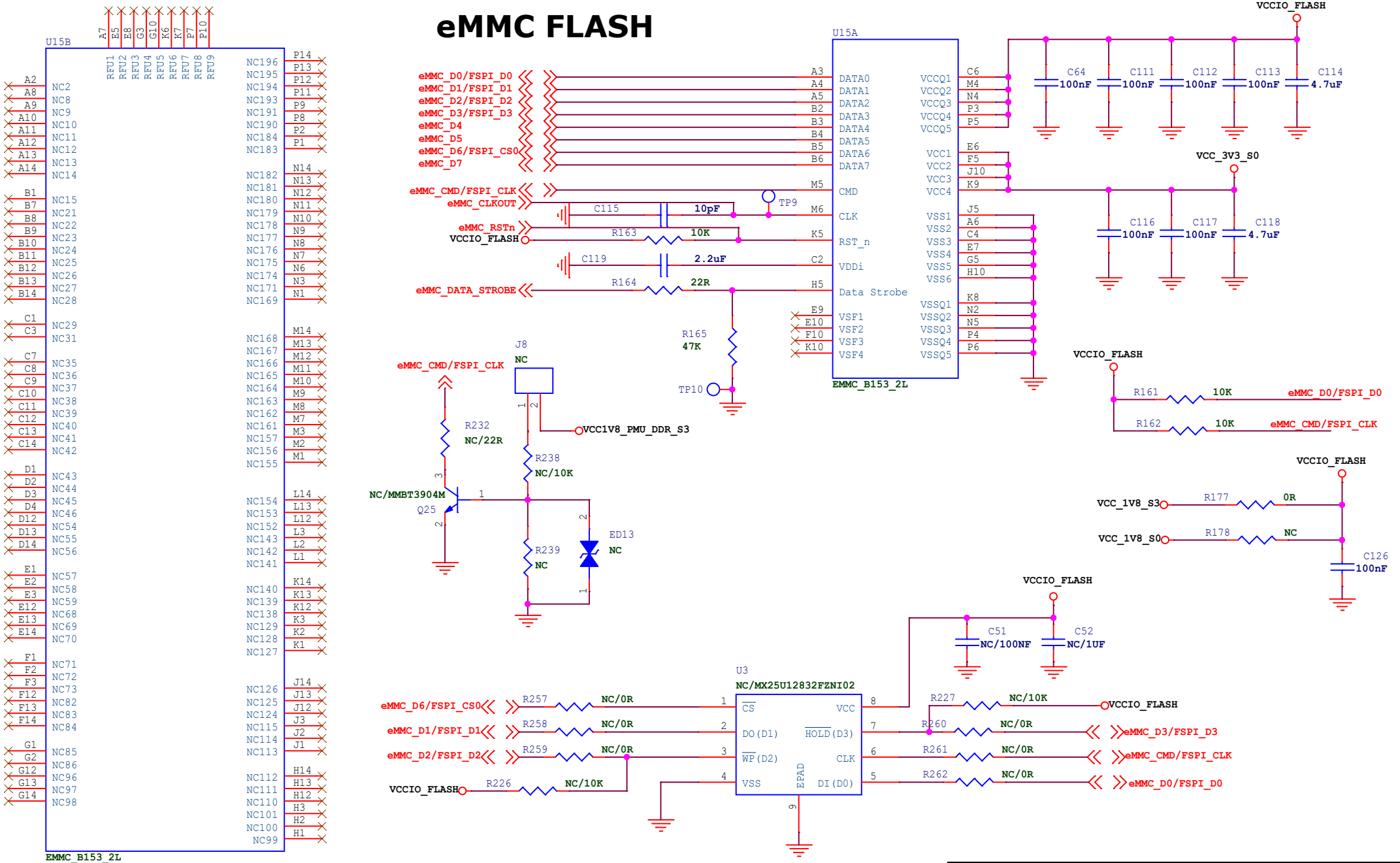




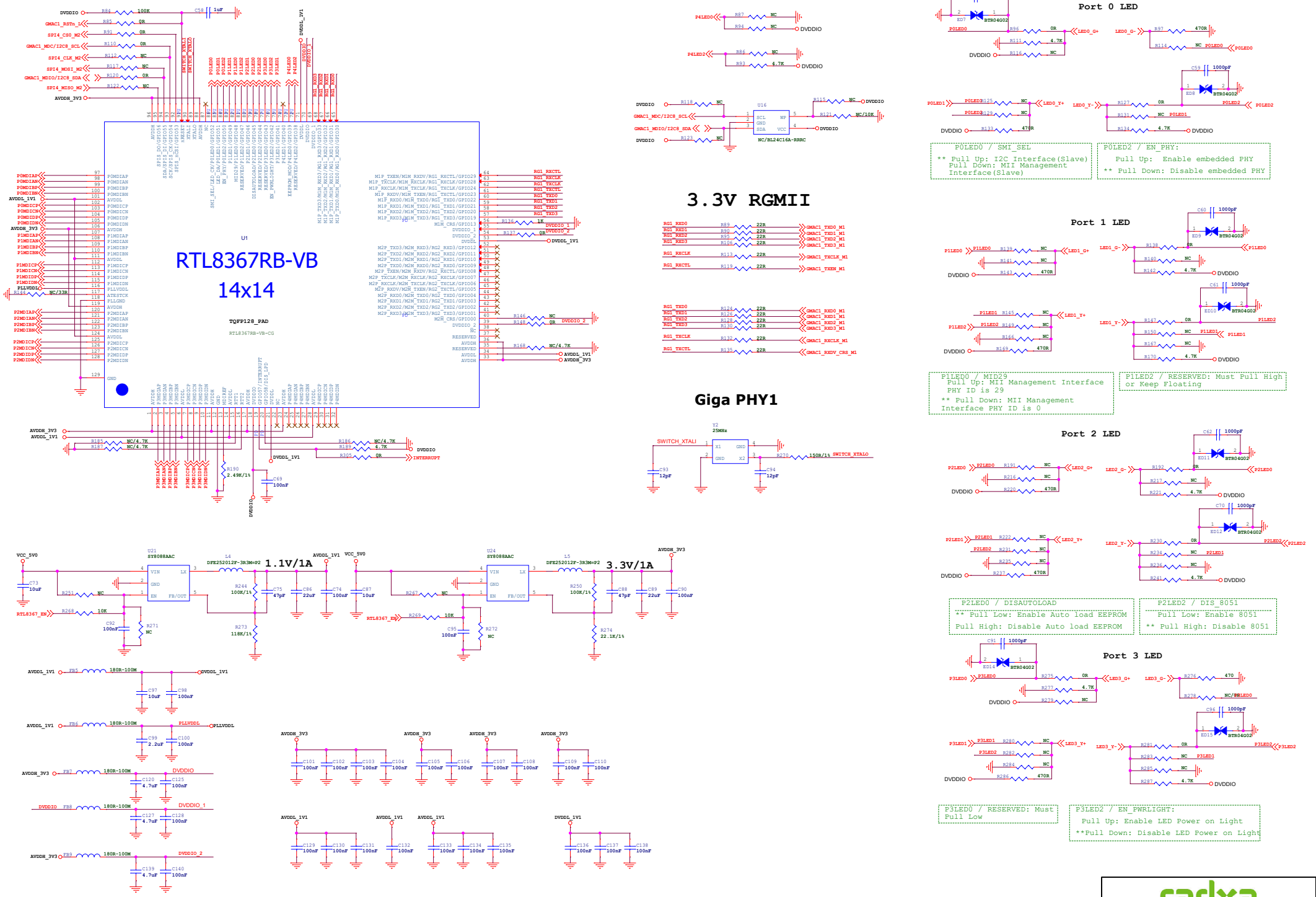


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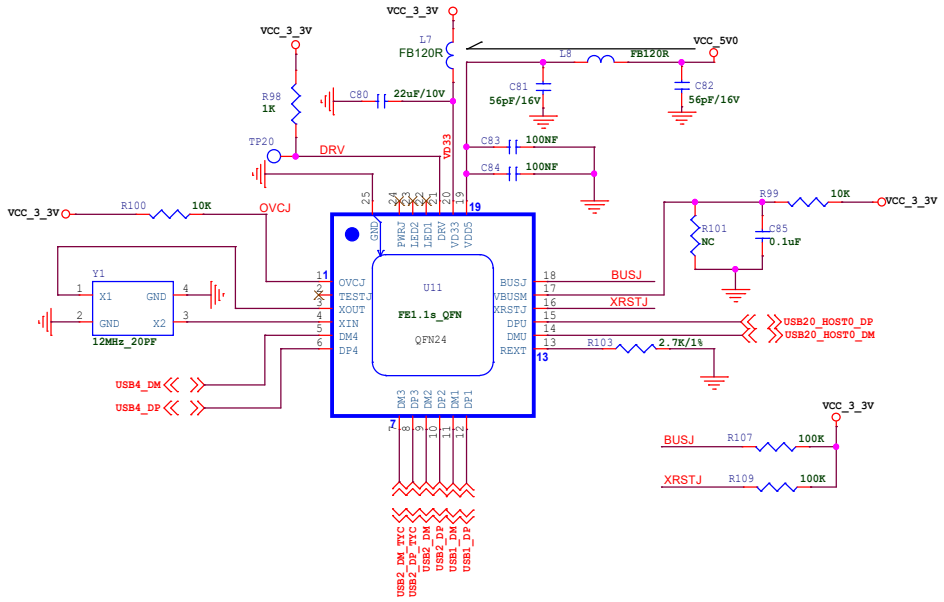
eMMC FLASH

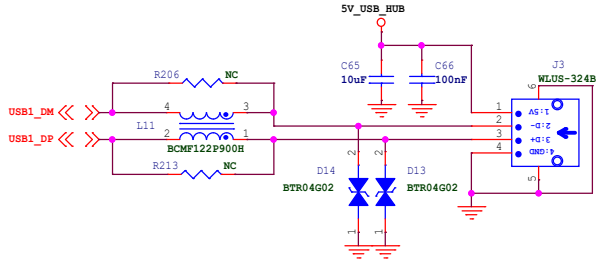


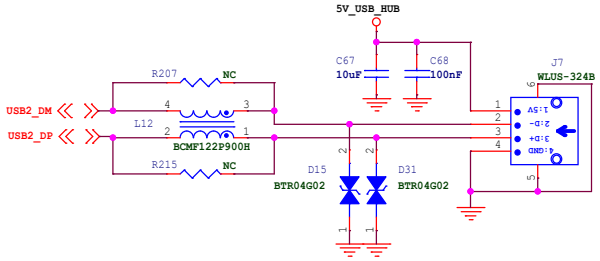
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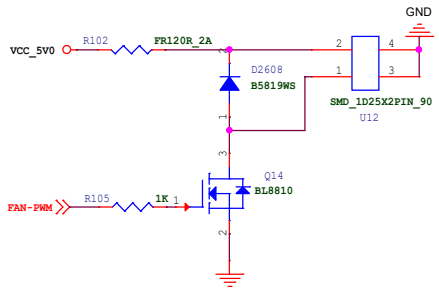

RTL8367RB-VB
14x14
TQFP128_PAD
RTL8367RB-VB-CG
3.3V RGMII
Giga PHY1
1.1V/1A
3.3V/1A
SY8088AAC
DFE252012F-3R3M=P2
NC/BL24C16A-RRRC
Port 0 LED
Port 1 LED
Port 2 LED
Port 3 LED
P0LED0 / SMI_SEL
** Pull Up: I2C Interface(Slave)
Pull Down: MII Management Interface(Slave)
P0LED2 / EN_PHY:
Pull Up: Enable embedded PHY
** Pull Down: Disable embedded PHY
P1LED0 / MID29
Pull Up: MII Management Interface PHY ID is 29
** Pull Down: MII Management Interface PHY ID is 0
P1LED2 / RESERVED: Must Pull High or Keep Floating
P2LED0 / DISAUTOLOAD
** Pull Low: Enable Auto load EEPROM
Pull High: Disable Auto load EEPROM
P2LED2 / DIS_8051
Pull Low: Enable 8051
** Pull High: Disable 8051
P3LED0 / RESERVED: Must Pull Low
P3LED2 / EN_PWRLIGHT:
Pull Up: Enable LED Power on Light
**Pull Down: Disable LED Power on Light
radxa
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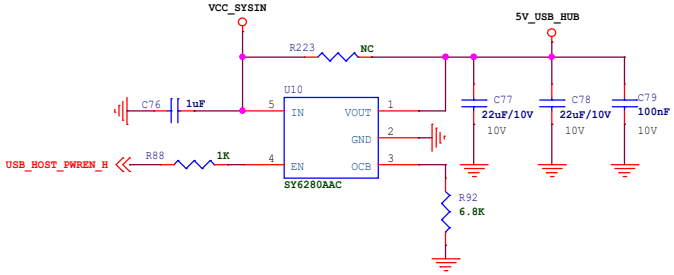

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P0MDIDP
P0MDICN
P0MDICP
P0MDIBN
P0MDIBP
P0MDIAN
P0MDIAP
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FB15
FB16
FB17
BCMEF122P900H
J4
TGE-2401SR
MDI0_3-
MDI0_3+
MDI0_2-
MDI0_2+
MDI0_1-
MDI0_1+
MDI0_0-
MDI0_0+
R340
R341
R342
R343
75R
C192
C195
C196
C197
22nF/25V
C198
0.1uF/25V
C199
1nF/2KV
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P1MDIDP
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P1MDICP
P1MDIBN
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P1MDIAN
P1MDIAP
FB1
FB2
FB3
FB4
J10
TGE-2401SR
MDI1_3-
MDI1_3+
MDI1_2-
MDI1_2+
MDI1_1-
MDI1_1+
MDI1_0-
MDI1_0+
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R345
R346
R347
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C203
C204
C205
C206
C207
RJ45-2
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P2MDIDP
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P2MDICP
P2MDIBN
P2MDIBP
P2MDIAN
P2MDIAP
FB22
FB23
FB24
FB25
J12
TGE-2401SR
MDI2_3-
MDI2_3+
MDI2_2-
MDI2_2+
MDI2_1-
MDI2_1+
MDI2_0-
MDI2_0+
R348
R349
R350
R351
C208
C211
C212
C213
C214
C215
RJ45-3
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P3MDIAP
FB18
FB19
FB20
FB21
J15
TGE-2401SR
MDI3_3-
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MDI3_2-
MDI3_2+
MDI3_1-
MDI3_1+
MDI3_0-
MDI3_0+
R352
R353
R354
R355
C216
C219
C220
C221
C222
C223
J5
HR5622S-118801B-21
MDI0_0+
MDI0_0-
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MDI0_1-
MDI0_2+
MDI0_2-
MDI0_3+
MDI0_3-
LEFT
RIGHT
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C194
NC
0R
LED0_Y+
LED0_Y-
LED0_G+
LED0_G-
J11
HR5622S-118801B-21
MDI1_0+
MDI1_0-
MDI1_1+
MDI1_1-
MDI1_2+
MDI1_2-
MDI1_3+
MDI1_3-
LEFT
RIGHT
RJ45_LED_TH12_LIE
C201
C202
NC
0R
LED1_Y+
LED1_Y-
LED1_G+
LED1_G-
J14
HR5622S-118801B-21
MDI2_0+
MDI2_0-
MDI2_1+
MDI2_1-
MDI2_2+
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MDI2_3+
MDI2_3-
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RIGHT
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C209
C210
NC
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LED2_Y+
LED2_Y-
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LED2_G-
J18
HR5622S-118801B-21
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MDI3_1-
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LED3_G+
LED3_G-
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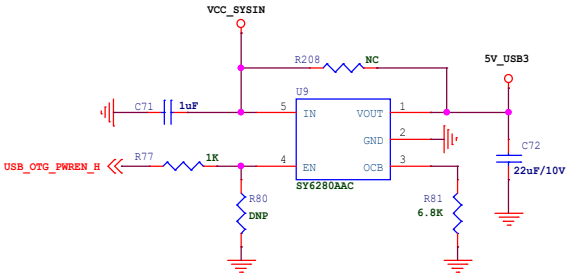


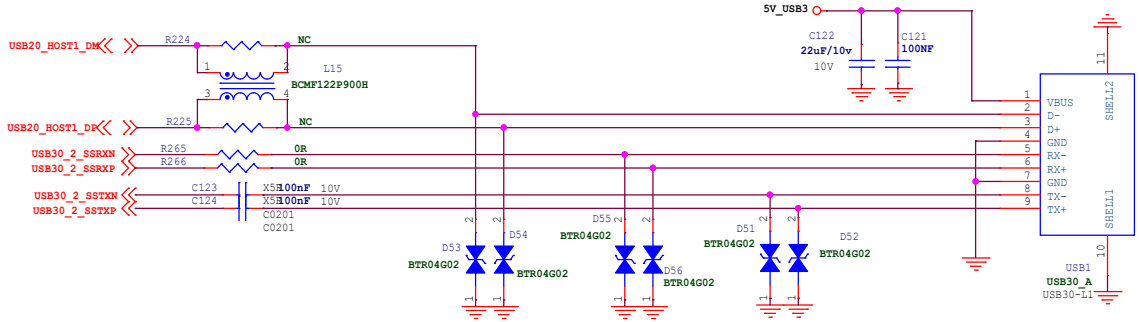






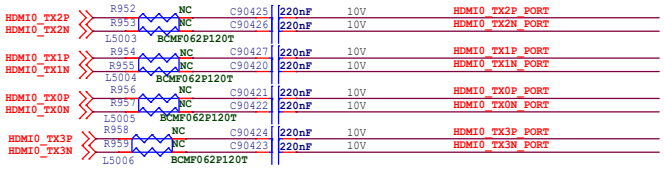


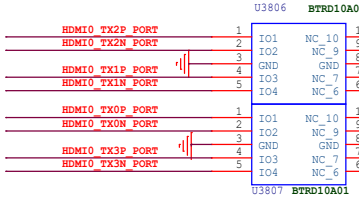


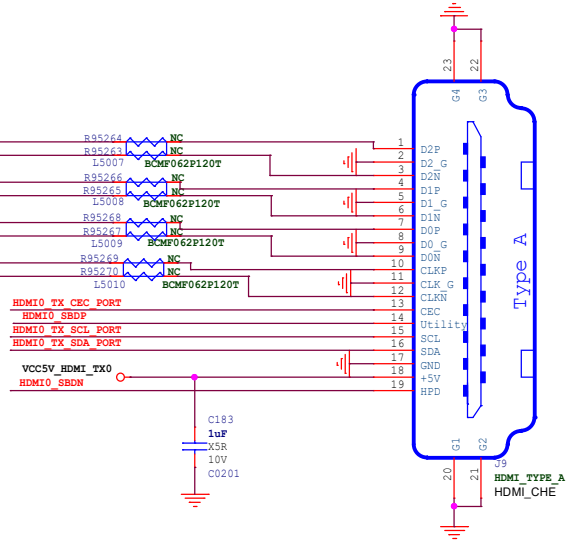


Size	Title:	E54C	REV
A3	Page Name:	USB HUB/FAN/USB Por	V1.2
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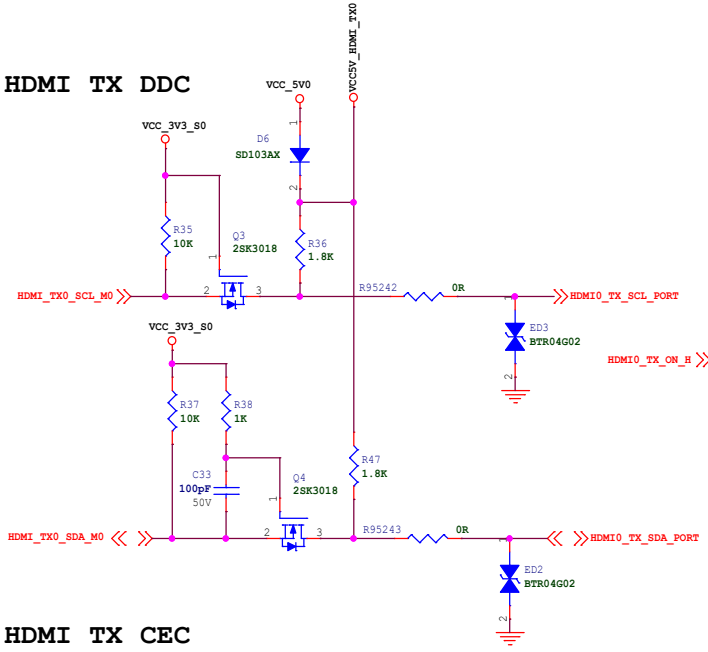
HDMI2.1 TX0

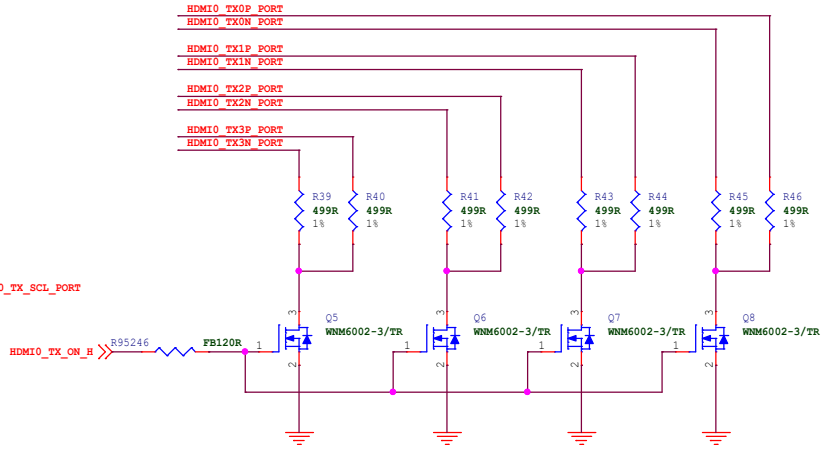




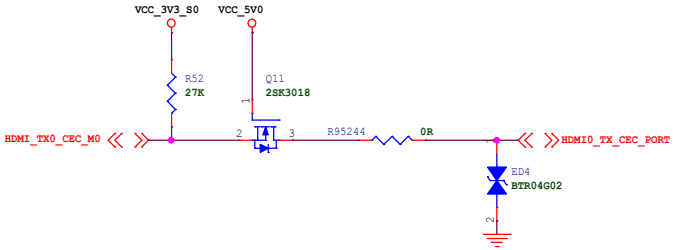


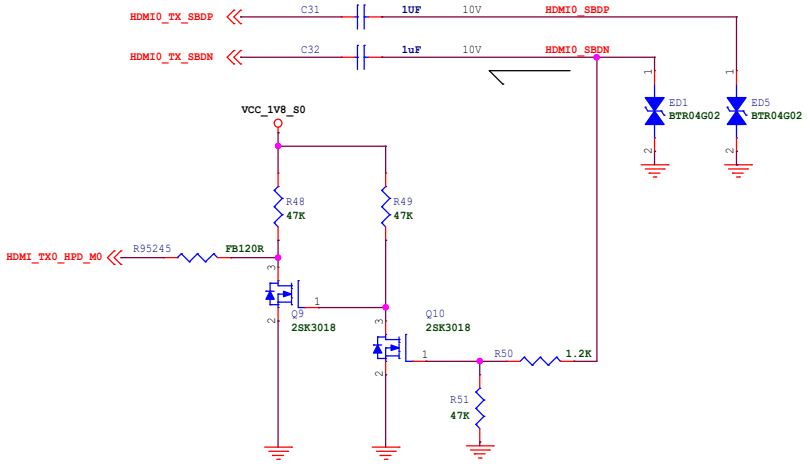
HDMI TX DDC





HDMI TX CEC

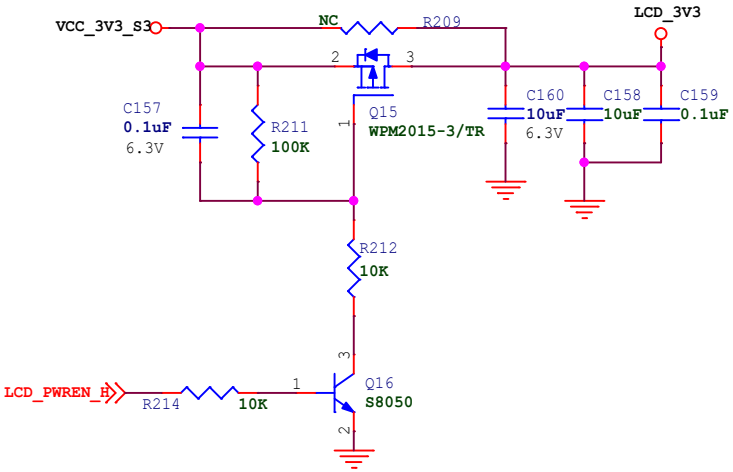


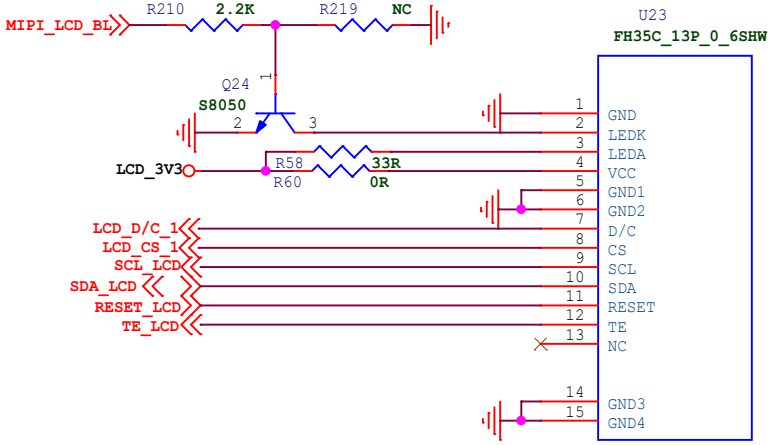


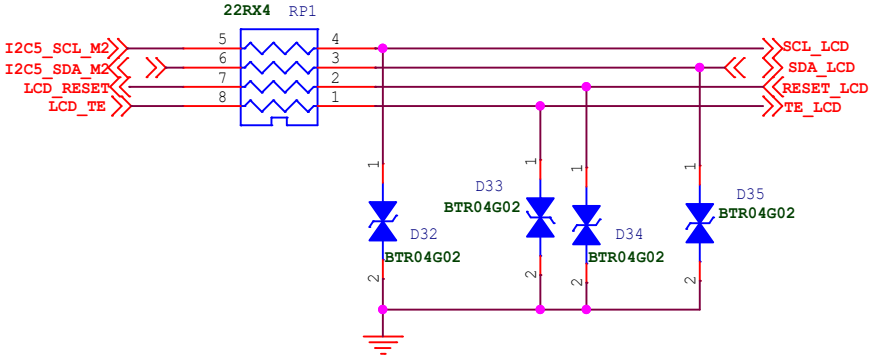
radxa

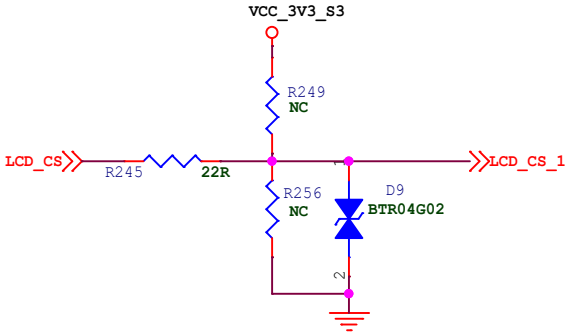
Size	Title:	E54C	REV
A3	Page Name:	HDMI 8K	V1.2
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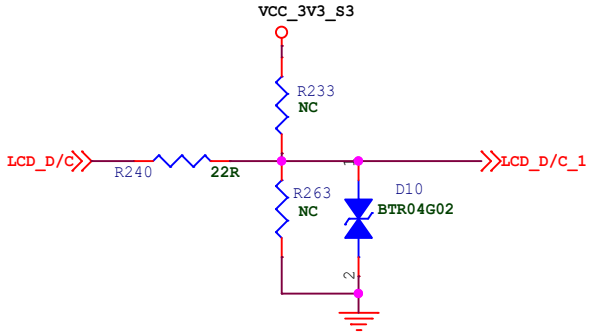
TFT





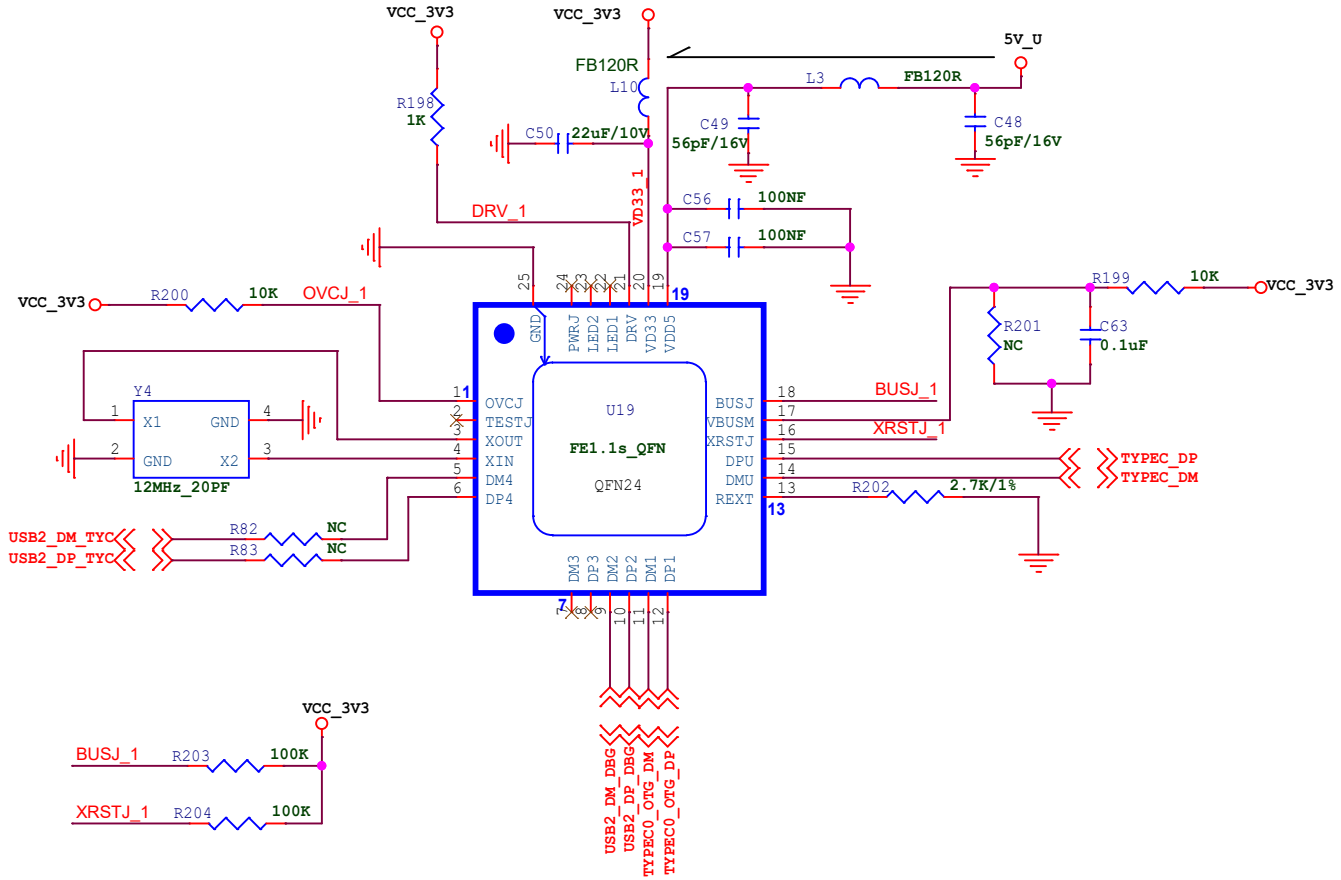


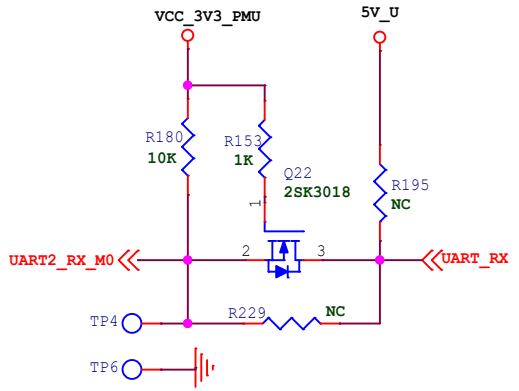


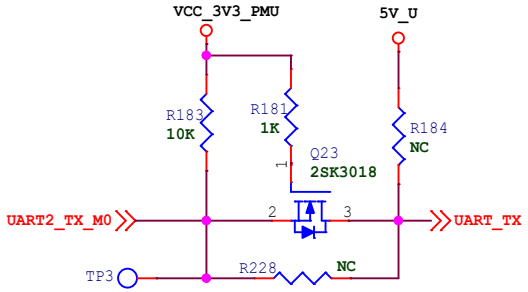


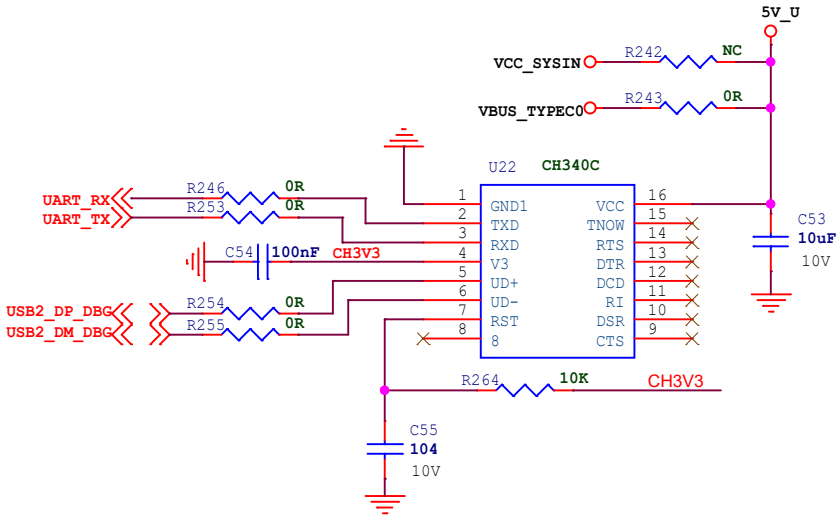
radxa

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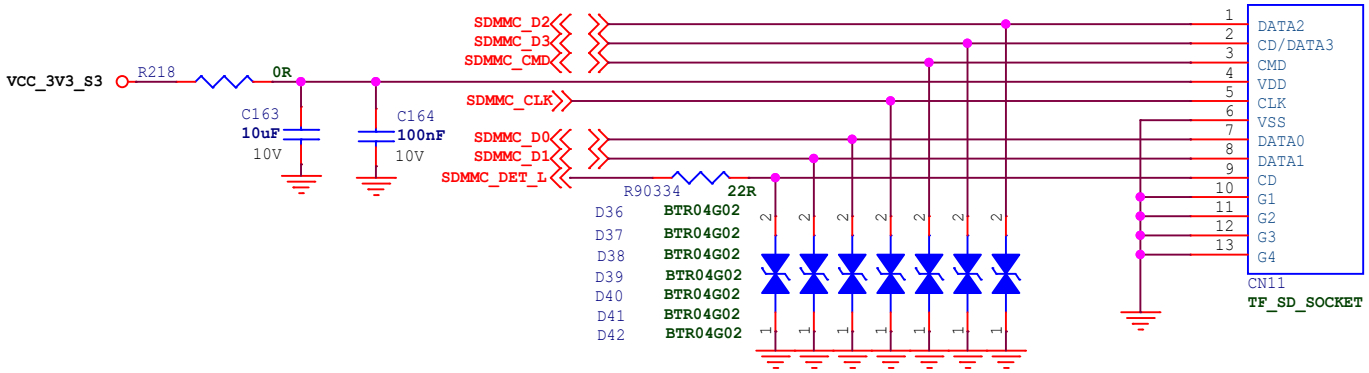




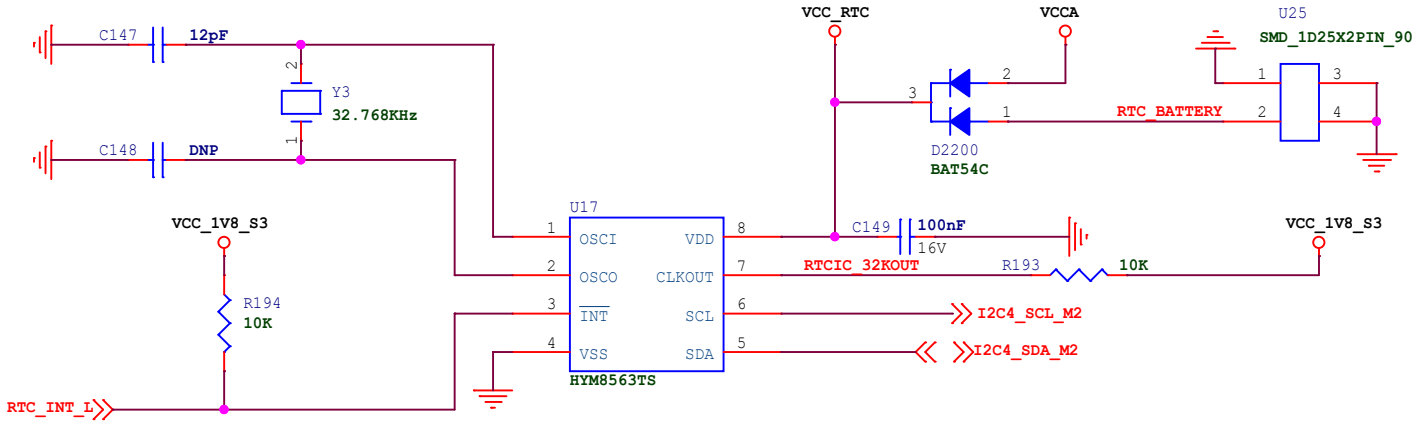


Size	Title:	E54C	REV
A4	Page Name:	UART/USB HUB	V1.2
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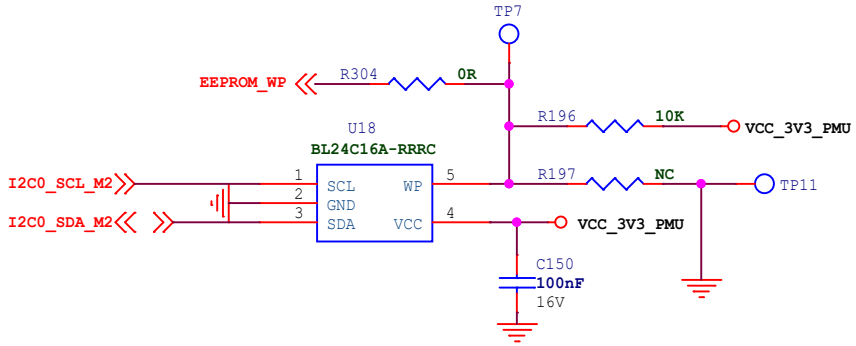
MicroSD Card



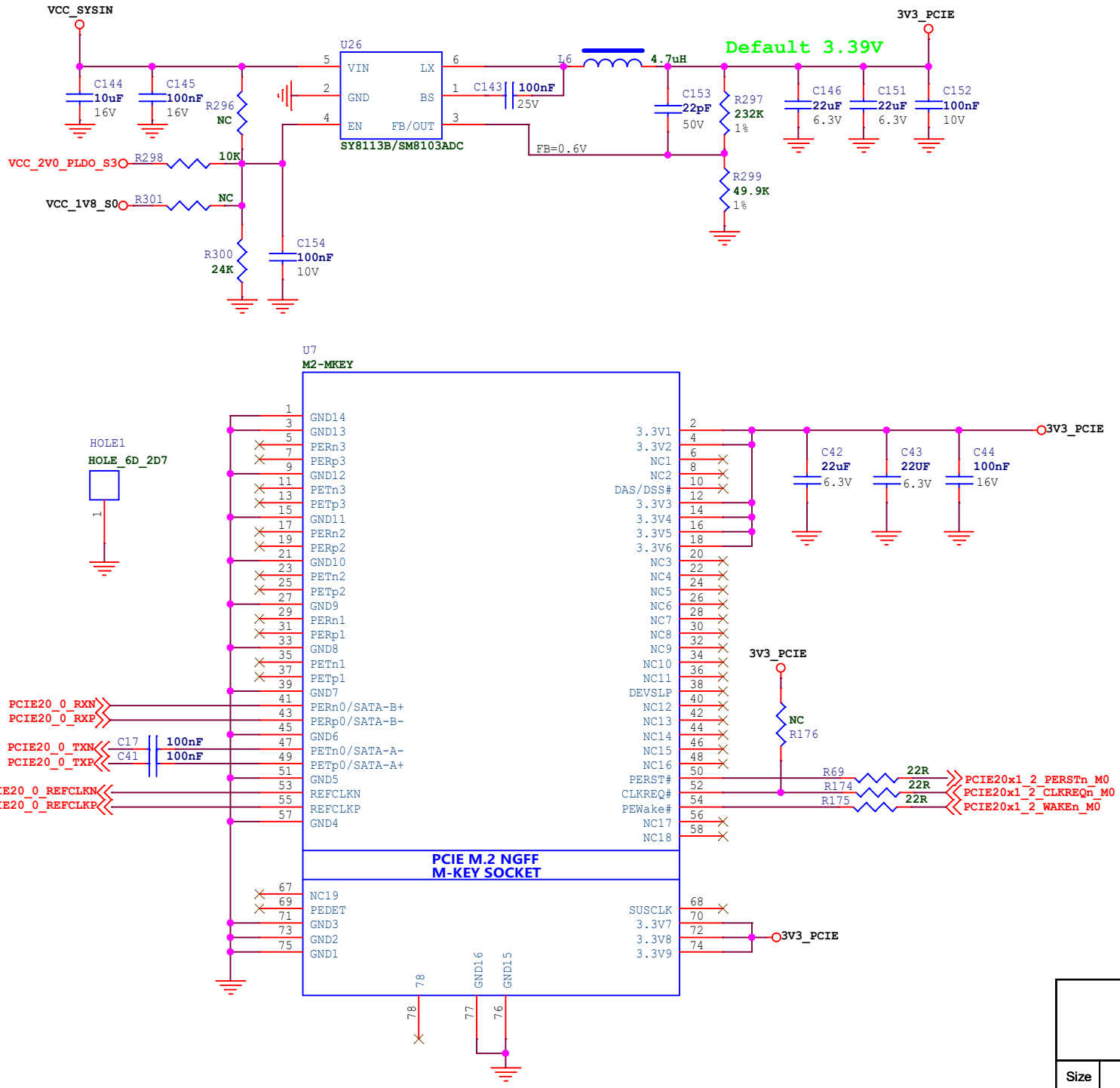
RTC IC



EEPROM

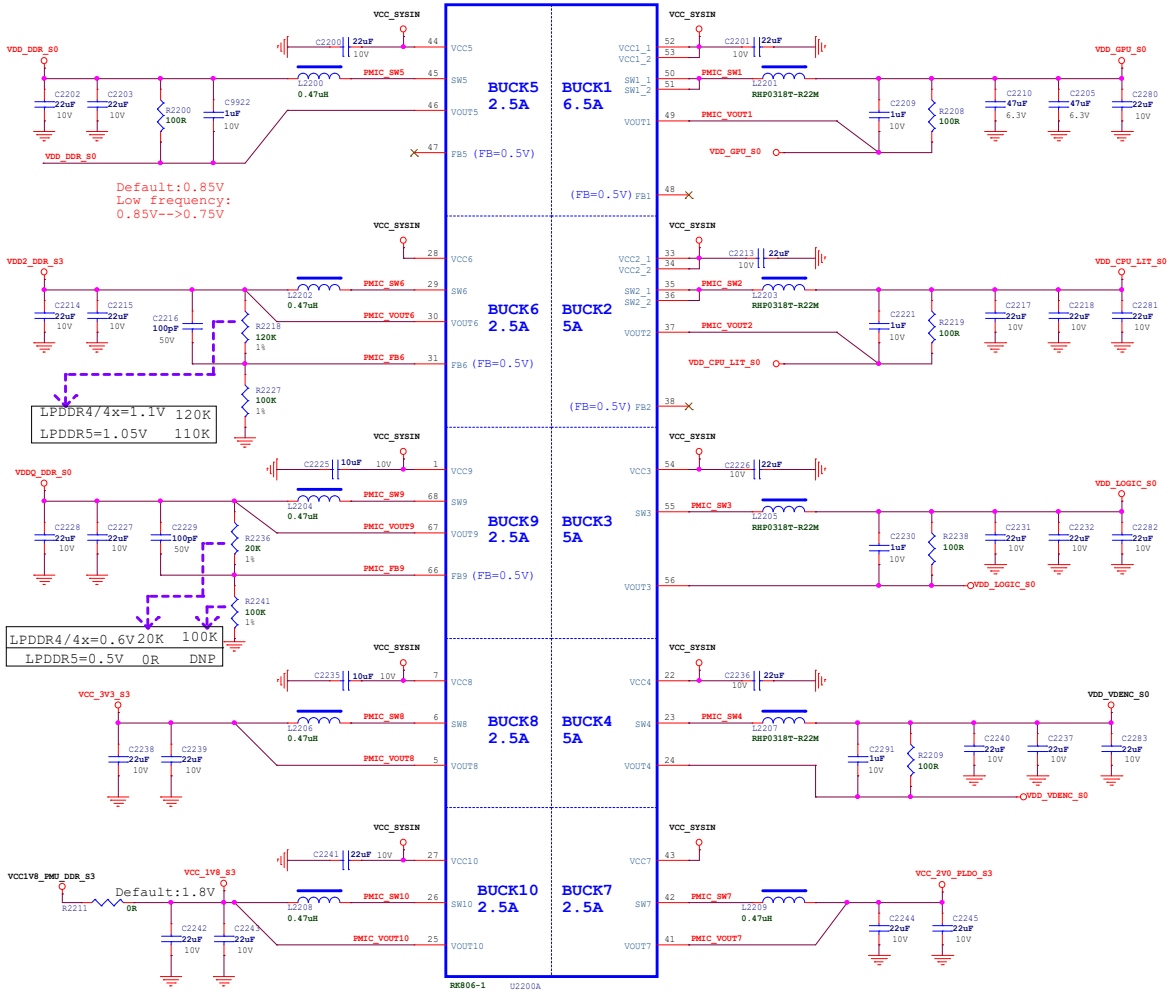


Size	Title:	E54C	REV
A4	Page Name:	TF/RTC/EEPROM	V1.2
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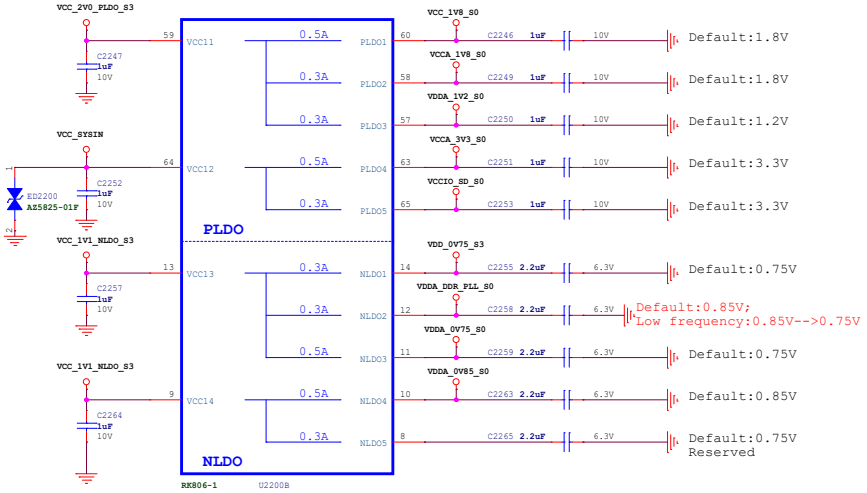


Size	Title:	E54C	REV
A4	Page Name:	M Key	V1.2
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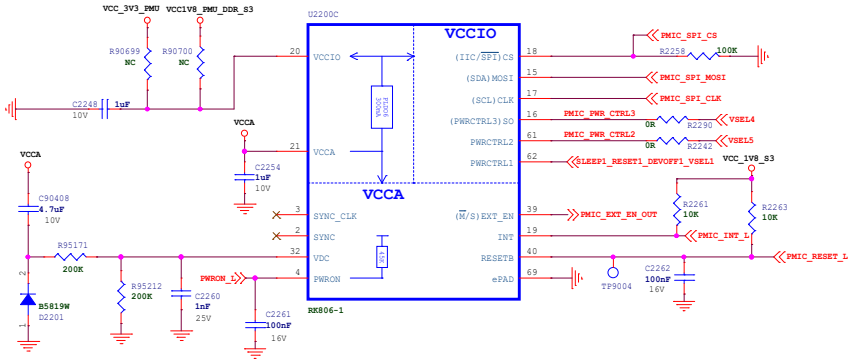
PMIC RK806-1 BUCK



PMIC RK806-1 LDO

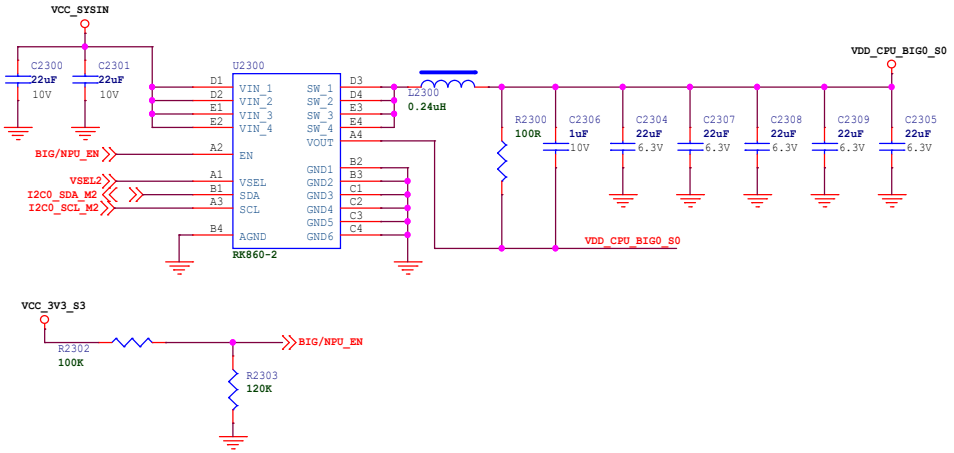


PMIC RK806-1 Managerment

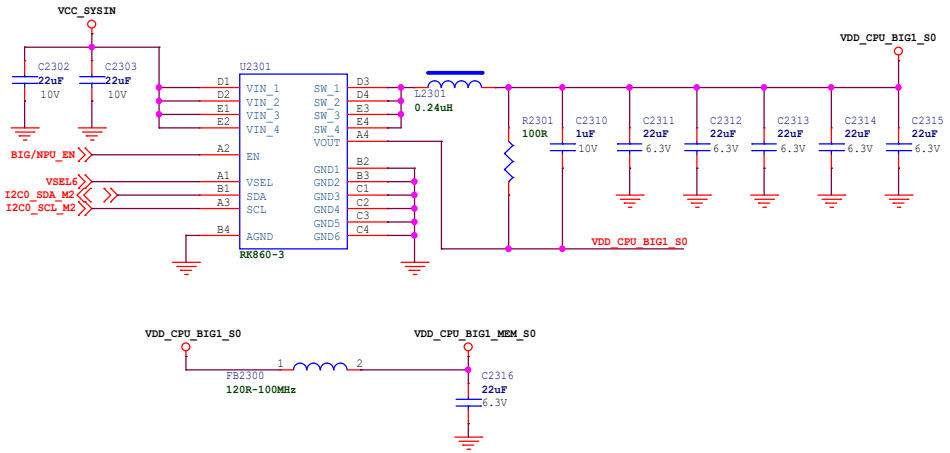


Size	Title: E54C		REV
Custom	Page Name: Power_PMIC		V1.2
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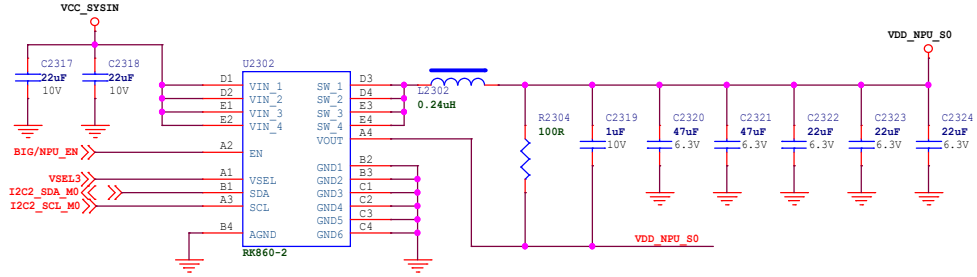
VDD_CPU_BIG0



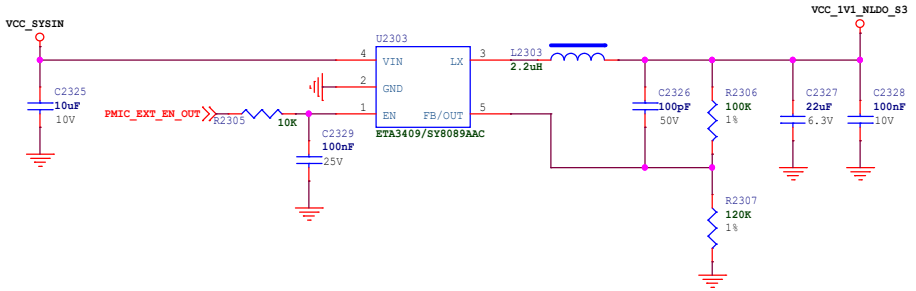
VDD_CPU_BIG1



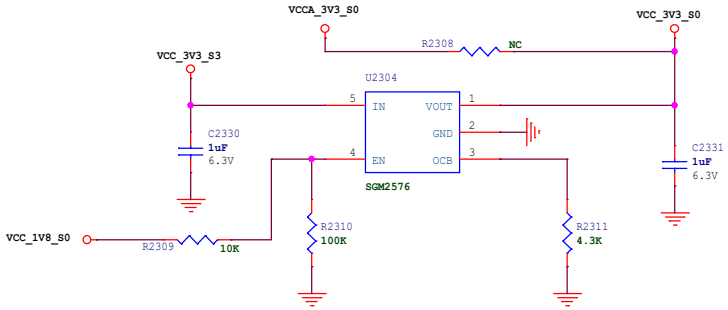
VDD_NPU



VCC_1V1_NLDO

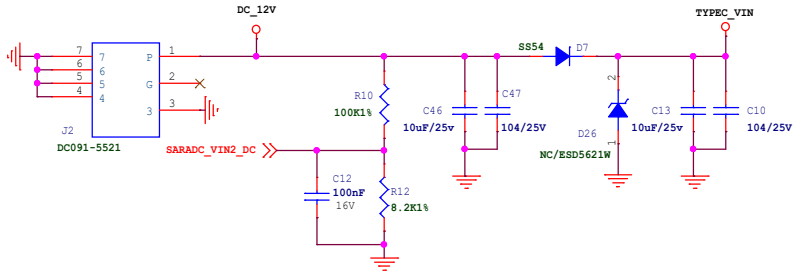


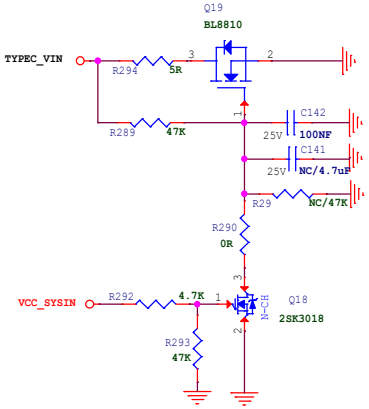
VCC_3V3_S0

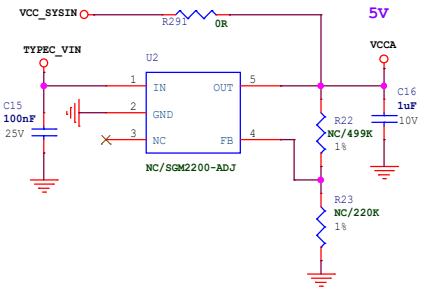


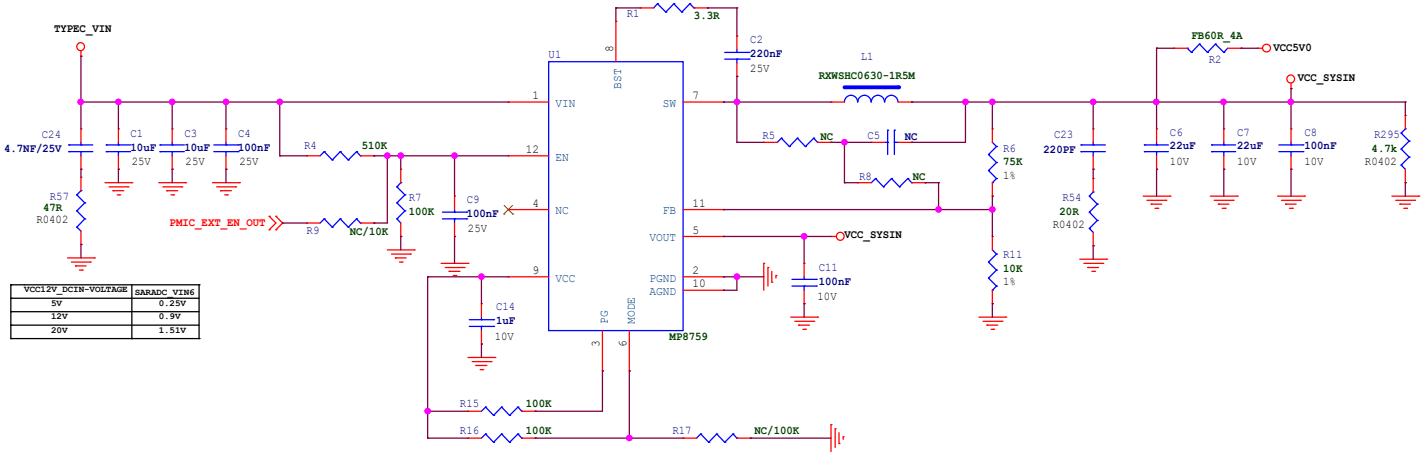
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DC12V



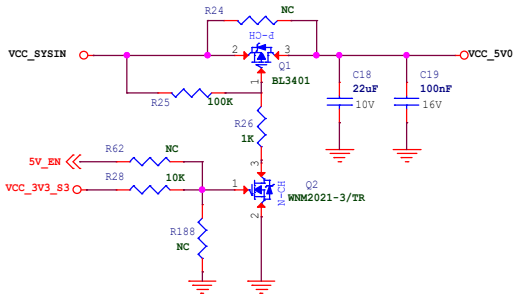






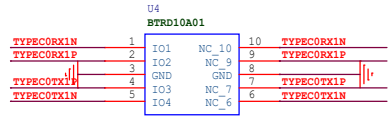
VCC12V_DCIN-VOLTAGE	SARADC_VIN6
5V	0.25V
12V	0.9V
20V	1.51V

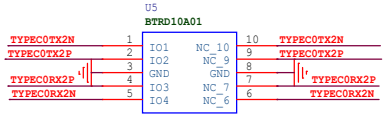
VCC_5V0

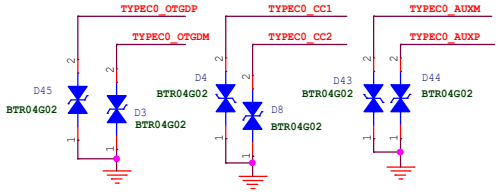


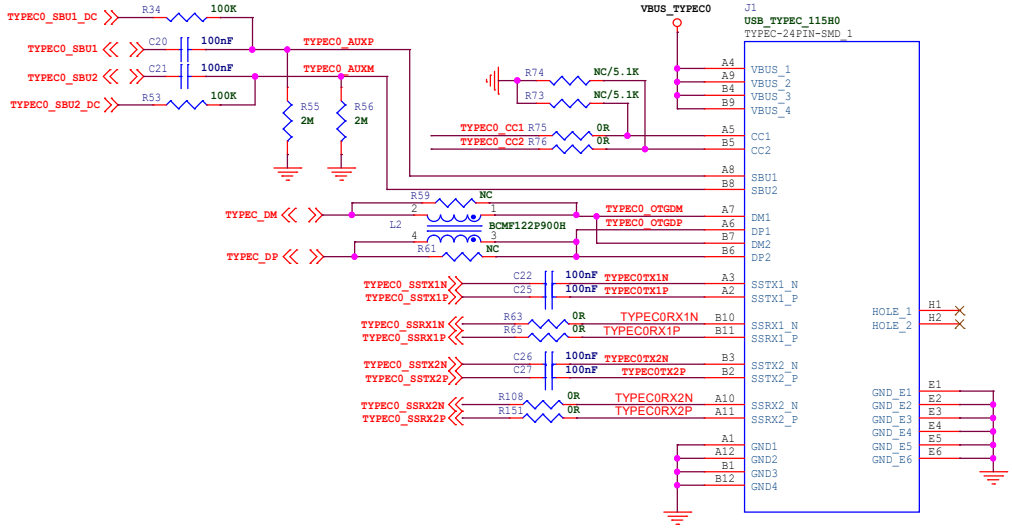
Size: A3	Title: E54C		REV: V1.2
	Page Name: Power_DC IN		
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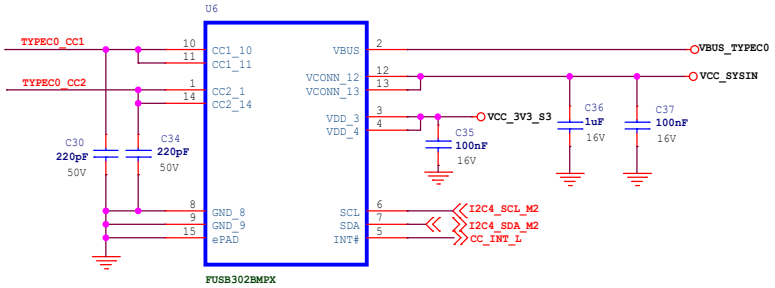
Type-C PORT

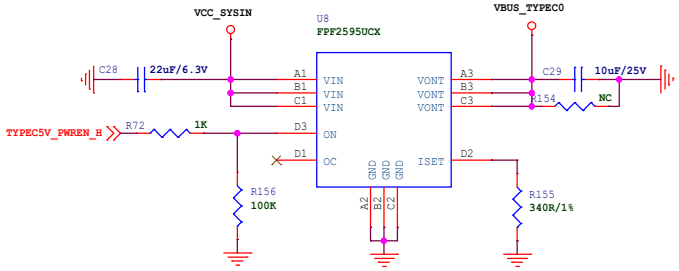




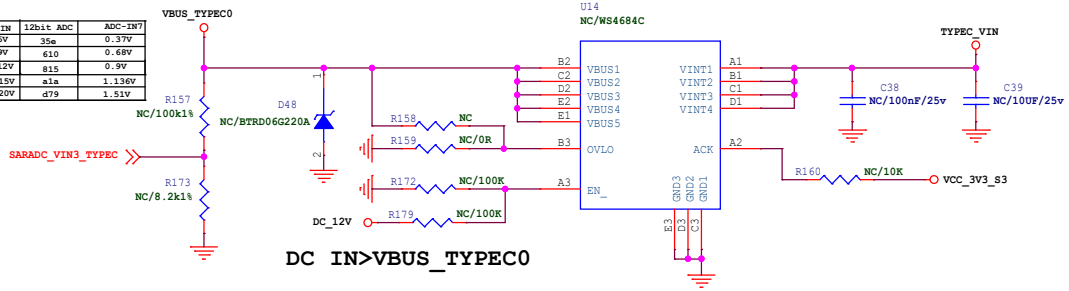


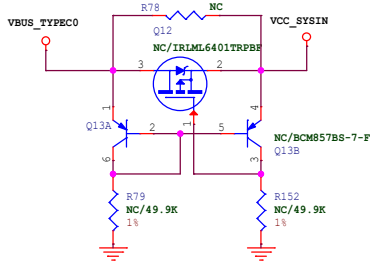






TYPEC_IN	12bit ADC	ADC-IN7
5V	35e	0.37V
9V	610	0.68V
12V	815	0.9V
15V	a1a	1.136V
20V	d79	1.51V

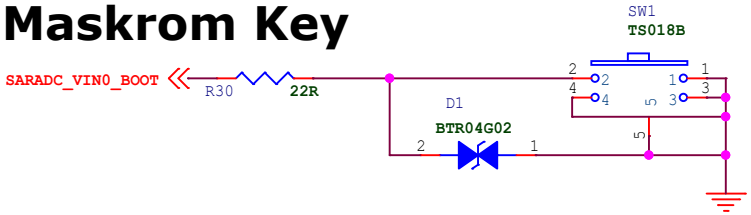




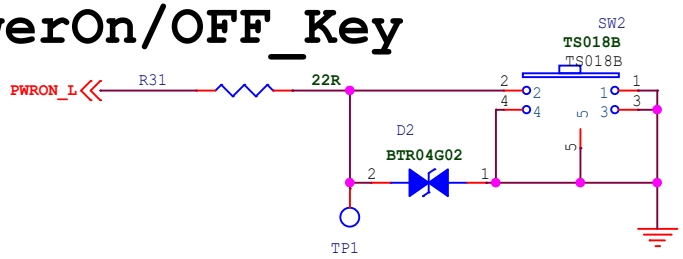
radxa

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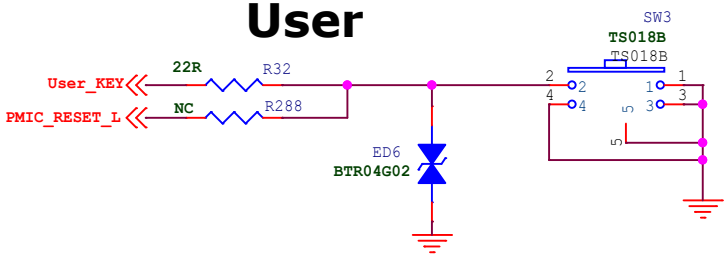
Maskrom Key

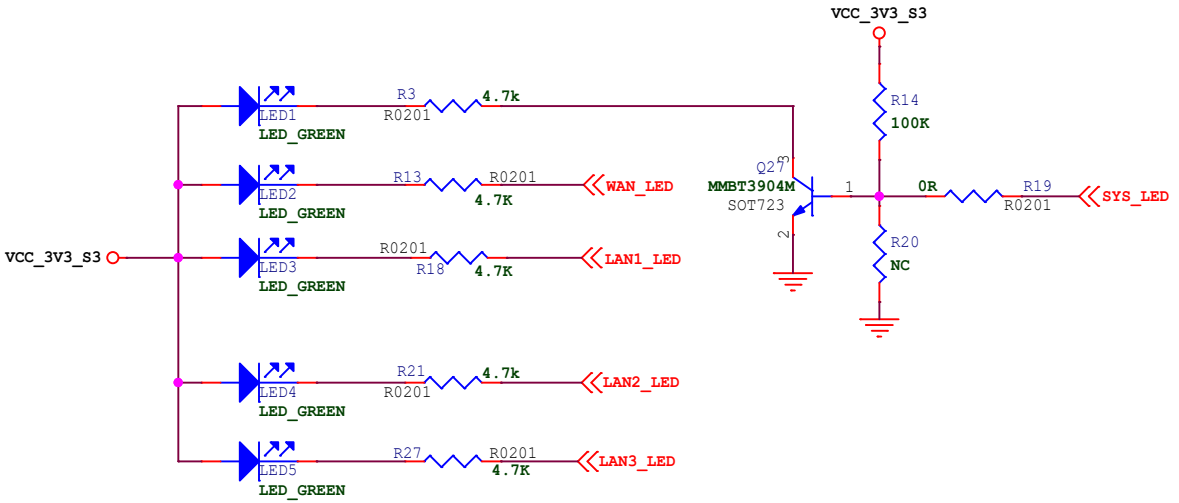


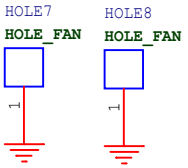
PowerOn/OFF_Key

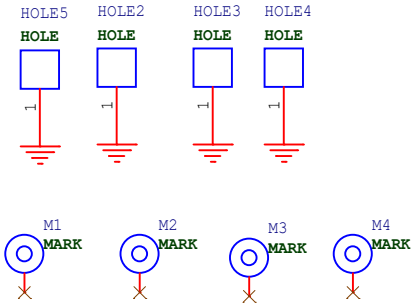


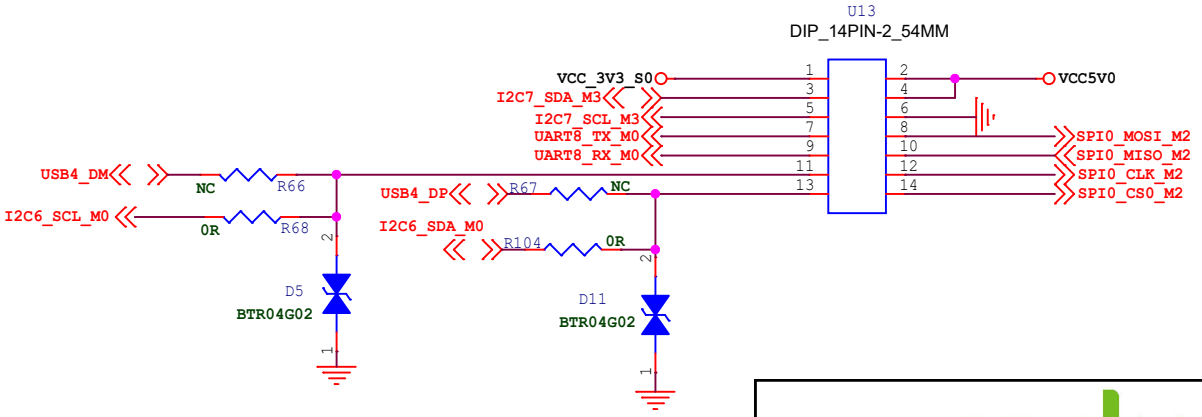
User











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